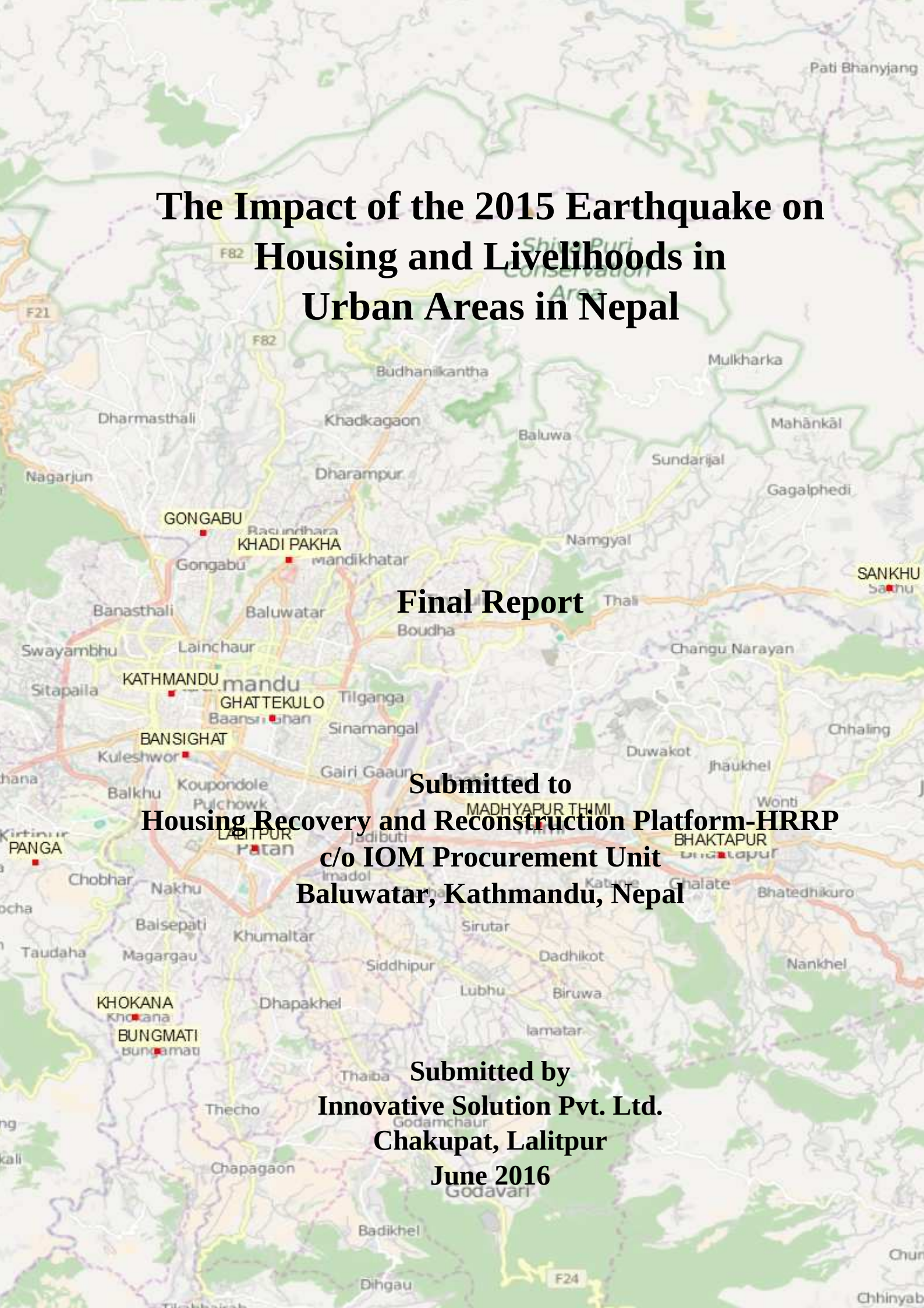




The Impact of the 2015 Earthquake on Housing and Livelihoods in Urban Areas in Nepal

Final Report

**Submitted to
Housing Recovery and Reconstruction Platform-HRRP
c/o IOM Procurement Unit
Baluwatar, Kathmandu, Nepal**

**Submitted by
Innovative Solution Pvt. Ltd.
Chakupat, Lalitpur
June 2016**

Study Team

Bhushan R Shrestha, PhD	Team Leader / Urban Information Management Specialist
Bijaya Krishna Shrestha, PhD	Senior Urban / Infrastructure Planner
Kishore L Manandhar	Senior Urban Planner
Kabita B Shrestha, PhD	Data Analysis/ Evaluation Expert
Sagar Joshi	Structural Engineer
Prakash Darnal	Cultural & Heritage Expert
Dibyeswar Shrestha	Socio-economist
Roshan B Shakya	Financial Analyst
Bhushan Bhandary	GIS Analyst
Madhav Adhikari	Field Coordinator
Mamata Sayami	Field Coordinator
Urmila Maharjan	Field Coordinator

Executive Summary

The M7.8 rector Gorkha earthquake of 25th April, 2015 and major subsequent aftershocks caused total death of 8,898 person and injured 22,309 person with destruction of 604,930 houses and damages of 288,856 houses in Nepal. According to PDNA, a total estimated loss is of NPR 706,461million (US\$ 706 million). The distribution of disaster effects between private and public sector shows more than three-fold damages in the private sector compared to the public sector.

Most post-disaster damage assessments have generally failed to record the impact of the disaster on the more vulnerable communities - the urban poor in general plus urban renters (particularly those living in frequently overcrowded conditions) and other groups that have suffered most. The impact of the disaster on these groups is further increased as the coping strategies they have are limited because of their pre-earthquake situations, viz., lack of family or community support mechanisms, limited access to alternative means of sheltering themselves, and overall lack of financial resources to bide them over their rough times.

Acknowledging this fact, UN-Habitat, IOM and Partner Organizations (POs) that are members of the HRRP has recognized the need to support the recovery and reconstruction efforts being undertaken in urban settings. Within the framework of the special Urban Recovery Coordination Group (URCG), HRRP was set to convene urban actors, conduct urban assessments, map out actions plans and encourage POs to develop actions appropriate within an urban recovery context.

The main objective of the study is to analyse the impact of the disaster on housing and livelihood including culture and heritage at seven major sectors of urban areas - Kathmandu valley core, peri-urban area, densely settled area, historic towns in fringes, market towns out of valley, informal settlements, and apartment buildings- thereby helping to the development of an urban housing recovery and reconstruction policy. To fulfil this main theme, it not only intends to document the earthquake impact on urban communities' housing conditions and livelihoods in a selected number of urban centres and their coping mechanism but also to assess the earthquake response by municipal authorities, urban communities, CBOs and CSOs and support agencies including analysis of urban communities' needs and challenges to move forward.

This study uses multi-sited case study approach based on primary and secondary data and information. The primary data and information were collected through Key Informants Interviews (KII) with municipal authorities and other relevant government authorities and individuals, Focus Group Discussion (FGD) with stratified group of people, and Technical Engineering Survey, and direct field observation of selected earthquake affected settlements. Specific focus was given to perspectives of local stakeholders such as communities, municipal authorities, and real victims.

Critical review and analysis of the data reveals that there are varying degree of impacts on housing and livelihood of these urban areas. Human factor through structural process has played a decisive role than natural phenomenon itself in damage and losses by the earthquake. In the core and historic towns in the fringe areas, majority of old houses built with unreinforced brick masonry where either completely collapsed or damaged due to vertical division and haphazard renovation and reconstruction, lack of timely maintenance, poor strength of binding material, and absence of tie beam around the house. Earthquake victims residing in these houses

have been suffered from loss of job and rent, displaced to places and above all residing either in temporary shelters or staying in the vulnerable damaged houses. Modern houses of RCC in the peri-urban areas as well as in the market towns have been damaged. The multiple reasons of destruction include poor construction quality, inadequate placement of rod, and illegal construction. They have also suffered from loss of their business and income, inadequate food supply and paying high cost for rent. Those living in the apartments also displaced as the monitoring and evaluation of the structural integrity is underway. Though there is no visible damage on informal settlements, nonetheless, they also suffered from the loss of job and income. Poor renters are one of the most vulnerable groups living in unhygienic and unsafe buildings.

GON, local municipalities together with various donor agencies including local communities, NGOs and CBOs actively participated in rescue and relief operation after the earthquake. They have not only distributed the foods and cloths but also assisted in construction of temporary shelters at various locations. In addition to these, the GON has come up with directives for safer construction, decided to provide either NPR 200,000 as grant or soft loan of NPR 1500,000 – NPR 2500,000. Moreover, the NRA has come up with loan and grant distribution mechanism, has established project implementation units at central and district levels. Many municipalities have revised their building bylaws as per GON's directives to ensure safer construction in post-earthquake period. Despite all, there are still numerous issues that need to be addressed. Many municipalities are yet to manage the debris of the damaged houses particularly located in the inner lanes and courtyards. Dismantling the damage houses has possessed another problem in the core area, where most of the houses are jointly built with common walls. Conservation of the past cultural heritages (tangible and non-tangible), safer and cost effective reconstruction are the major issues to be solved in the reconstruction plan and process. Moreover, the new development being an opportunity shall also incorporate the components of Green Homes and energy efficiency principles, besides the safety and cost effectiveness. As there is no clear cut vision and strategy for the reconstruction and recovery process for the urban areas, many communities and groups have come up with their own proposal for reconstruction with little knowledge on the implementation techniques. Reconstruction plan based on the concept of house pooling will be difficult to implement within the framework of the existing legal and institutional framework. Lack of finance and confusion over the government policy towards the reconstruction are the two major issues identified by the communities for delay in reconstruction work. Delay in reconstruction process may lead to destruction of traditional 'newari' towns of the historic core areas and in the fringe area thereby negatively affecting the local economy, generated by the tourism. Moreover, the chance of conversion of the existing temporary shelters at various locations into slums thereby forming new squatter settlement is high.

To prevent the likelihood of such a chaotic urban future, the study has recommended "The Way Forward" for building back better (BBB) housing and livelihood of urban areas of the country. For that an integrated and holistic approach by combining buildings with infrastructure provision and physical redevelopment with livelihood enhancement is suggested. Active participation of business entrepreneurs, infrastructure providing agencies, and financial institutions is essential besides government agencies and local communities. As the local context of the damaged area, community background and their socio-cultural dimension and business opportunity are different, it is necessary to carry out the feasibility study regarding mode of redevelopment. Formulation of urban design guidelines to protect the shared community values as well as inclusion of energy efficient component is essential, which can be linked with planning and building permit system and incentive mechanism. A financial

model that also include household awareness should be developed so that each household will feel comfortable in contribution towards joint construction, and loan repayment.

Poor renters living in earthquake affected areas are highly neglected group in this earthquake disaster and due to the lack of data and information their status after earthquake is not clear. There is need of a comprehensive study to know their status to address this group by Government and other agencies working for their betterment. Similarly, people living in the informal settlements are not affected directly by this earthquake, but their issues have to be addressed by incorporating informal settlement management plan into government's normal planning process.

Table of Content

Contents

1	Introduction	1
1.1	Background	1
1.2	Objective	2
1.3	Scope	2
1.4	Rationale	2
1.5	Geographical coverage	2
2	Research Methodology	3
2.1	Information Requirement Analysis	3
2.1.1	Selection of Urban Settlements	4
2.1.2	Preparation of Base Map	4
2.1.3	Clustering of Settlement	4
2.1.4	Preparation of Checklist and Questionnaire	4
2.2	Collection of Data and Information	5
2.2.1	Preliminary Field Visit	5
2.2.2	Detailed Field Survey	5
2.2.3	Technical Survey	6
2.3	Analysis of Data and Information	6
3	Impact Assessment of Earthquake	8
3.1	Kathmandu Valley Core Settlement	11
3.1.1	Housing and Livelihood	11
3.1.2	Public Infrastructure	14
3.1.3	Culture and Heritage	14
3.1.4	Marginalised Group	14
3.2	Kathmandu Valley Peri-urban Settlement	15
3.2.1	Housing and Livelihood	16
3.2.2	Public Infrastructure	16
3.2.3	Culture and Heritage	16
3.3	Kathmandu Valley Densely Settled Area	17
3.3.1	Housing and Livelihood	17
3.3.2	Culture and Heritage	17
3.4	Kathmandu Valley Historical Towns in Fringe	18
3.4.1	Housing and Livelihood	18
3.4.2	Public Infrastructure	20
3.4.3	Culture and Heritage	20
3.4.4	Marginalised Group	20
3.5	Market Towns outside Kathmandu Valley	21
3.5.1	Housing and Livelihood	21
3.5.2	Public Infrastructure	23
3.5.3	Culture and Heritage	23
3.5.4	Marginalised Group	23
3.6	Informal Settlement	25
3.6.1	Housing and Livelihood	25
3.6.2	Cultural & heritage	25
3.7	Apartment Buildings	26
3.7.1	Housing and Livelihood	27
4	Rescue, Relief, Recovery and Reconstruction	29
4.1	Rescue Initiative	29

4.2	Relief Initiative	29
4.2.1	Government's Initial and Winterization Support.....	29
4.2.2	Ward Citizen Forum	30
4.2.3	DDRC and CDRC.....	30
4.2.4	Community Based Organizations & Local Youths	30
4.2.5	Other Organizations	31
4.3	Recovery Initiatives	32
4.3.1	Government's Financial Scheme for Recovery	32
4.3.2	Community's Perception in Traditional Buildings	33
4.3.3	Community's Interest in Planned Settlement	33
4.3.4	Government's Initiation in Safe Building Construction	34
4.3.5	Role and Capacity of Municipalities.....	34
4.4	Status of Reconstruction	35
4.4.1	Debris Clearance	37
4.4.2	Status of Undemolished Buildings.....	38
4.4.3	Safety Measures of Undemolished houses	39
4.4.4	Use of Damaged Houses	40
4.4.5	Restoration of Damaged Houses.....	41
4.5	Status of Buildings under Construction	41
4.5.1	Structure Type of Reconstructing Buildings.....	41
4.5.2	Building Level Category of Reconstructing Buildings.....	42
4.5.3	Column Size of the Reconstructing Buildings.....	42
4.5.4	Use of Trained Masons in Reconstructing.....	43
4.5.5	Purpose of Using Reconstructing Building.....	43
4.5.6	Building Permit Status of Buildings under Reconstruction.....	44
4.5.7	Soil Vulnerability of the Settlement	44
4.5.8	Other Technical Requirements of Reconstruction.....	44
5	Major Issues and Long Term Impacts	45
5.1	Major Issues	45
5.2	Long Term Impacts.....	52
6	The Way Forward.....	54

Bibliography

Annex

Annex 1. Profiles of Selected Urban Settlements

Annex 2.1. Selection Criteria of Urban Settlements

Annex 2.2. Number of Clusters and Major Characteristics

Annex 2.3. Checklist for Preliminary Field Visit

Annex 2.4. Semi-structured Questionnaire for FGD

Annex 2.5. Checklist for Technical Survey

Annex 2.6. List of Key Informa

Annex 2.7. Case Studies

Annex 3.1. Population, Households, Area of Urban Area of EQ affected Districts

Annex 3.2. Number of Fully Damaged Households of Urban Area of EQ affected Districts

Annex 3.3. Causes of Damage in Masonary Buildings

Annex 3.4. Information on Area, Population, Household and Damaged or corresponding
Municipalities of Selected Urban Settlement

Annex 4.1. Emerging Redevelopment Models

1 Introduction

1.1 Background

The Gorkha earthquake of 7.8 magnitude on 25 April 2015 and the strong aftershock on 12 May 2015 that struck the central and eastern regions of Nepal caused significant damage to private houses, public infrastructure, and heritage. According to the Ministry of Home Affairs (MoHA), over 604,930 houses were destroyed and 288,856 houses were damaged. According to PDNA, a total estimated damages (damages and losses) of NPR 706,461 million rupees (US\$ 7064million) in four major sectors such as social, productive, infrastructure and cross cutting issues and resulted in 8790 human casualties and injured 22,300 people in the country. The distribution of disaster effects between private and public sector shows more than three-fold damages in the private sector compared to the public sector. For instance, there is an estimated damages of NPR540, 362 million (US\$5,404million) in the private sector accounting for 76 percent as against NPR 166,100 million (US\$1,661 million) accounting for 24 percent. The loss in personal income due to damages in the productive sectors like agriculture, irrigation, industry, commerce, tourism, finance etc. is NPR17,124 million (\$171million).

To respond to these devastating earthquakes, the Nepal Shelter Cluster was activated by MoUD Government of Nepal (GON) and co-led by IFRC to coordinate the more than 120 agencies engaged in shelter activities. In October 2015, with the aim of providing a forum to agencies that are intending to engage in settlement and housing reconstruction the Recovery and Reconstruction Working Group (RRWG), co-led by IOM and UN-Habitat and endorsed by GON, was established with the objective to support and provide guidance to cluster members working in early recovery shelter interventions, to coordinate with government authorities and relevant stakeholders in the development of recovery and reconstruction policies and implementation strategies for the earthquake-affected areas. The **Housing Recovery and Reconstruction Platform (HRRP)** has taken over the functions of the RRWG from January 2016 with a focus on housing reconstruction.

UN-Habitat, IOM and Partner Organizations (POs) that are members of the HRRP has recognized the need to support the recovery and reconstruction efforts being undertaken in urban settings. Within the framework of the special Urban Recovery Coordination Group (URCG), HRRP was set to convene urban actors, conduct urban assessments, map out actions plans and encourage POs to develop actions appropriate within an urban recovery context.

Most post-disaster damage assessments have generally failed to record the impact of the disaster on the more vulnerable communities - the urban poor in general plus urban renters (particularly those living in frequently overcrowded conditions) and other groups that have suffered most. The impact of the disaster on these groups is further increased as the coping strategies they have are limited because of their pre-earthquake situations, viz., lack of family or community support mechanisms, limited access to alternative means of sheltering themselves, and overall lack of financial resources to bide them over their rough times. In this context, it is imperative to assess the impact of Gorkha earthquake on housing and livelihood of urban areas.

1.2 Objective

The main objective of the study is to analyse the impact of the disaster in the urban sector and thereby help to the development of an urban housing recovery and reconstruction policy. The specific objectives are as follows:

- i. Document the earthquake impact on urban communities' housing conditions and livelihoods in a selected number of urban centres and their coping mechanism;
- ii. Document the earthquake response in the selected urban centres with particular emphasis on those by municipal authorities, urban communities, CBOs and CSOs and support agencies;
- iii. Analyse urban communities' needs and challenges to move forward with recovery and reconstruction with particular emphasis on the vulnerable people (e.g. urban poor, poor renters, informal settlers, tenants, female headed households and elderly people).

1.3 Scope

The assessment includes all representative urban contexts affected by the 2015 earthquake: district centres, historic small towns, and affected areas of the Kathmandu Valley. 'Urban systems' will encompass: institutional, physical, social, cultural, economic systems. The scope of work include followings:

- Consolidate pre and post-earthquake data, develop a bibliography of data, references and resources.
- Map stakeholders for data collection, analysis and monitoring.
- Document post-earthquake humanitarian and reconstruction assistance, source, type, quantity, coverage.
- Develop post-earthquake situation analysis in order to inform policy and programme development and to develop a baseline in order to measure recovery progress.
- Develop proposals for continued monitoring: indicators, tools, capacities.

1.4 Rationale

Almost 30% of the losses estimated by PDNA is shared by urban areas. Apart from its rebuilding challenges urban areas are the service centres that need to be brought into shape to support the overall reconstruction works. The observation of the effects of earthquake shows that urban poor, particularly vulnerable population are hard hit by this disaster and there is a need to come up with policy tools and programmes to support them. On the other hand, the failure of the rural economy also brings migration and load to the urban economy. Furthermore, the complexities in urban society demand in-depth study and tailored tools to address the neediest populations within the urban society. In this context this study provides adequate evidences and areas of work that is needed for future planning of recovery and reconstruction works in urban Nepal.

1.5 Geographical coverage

Of the 14 districts severely affected by the earthquake the study has focused on seven municipalities within Kathmandu Valley and three market towns out of the valley - Chautara, Charikot and Gorkha. Based on the characteristics of the urban sub-groups, Kathmandu Valley further converges to city inner cores, peri-urban, city centres (i.e, densely settled), and historic towns on the fringes. The study also includes informal settlements and apartment buildings.

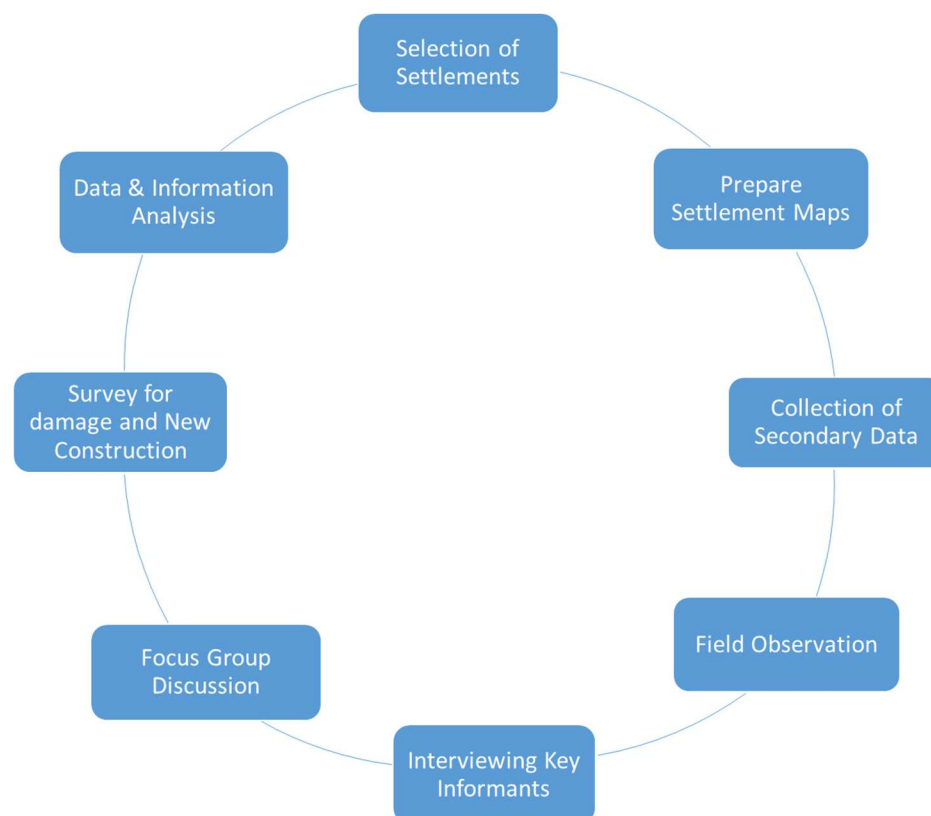
2 Research Methodology

This study has used a multi-sited case study approach based on primary and secondary data and information. The secondary data was collected from relevant districts, municipalities and agencies. The primary data and information were collected through:

- Key Informants Interviews (KIIs) with municipal authorities and other relevant government authorities and individuals.
- Focus Group Discussion (FGDs) with stratified group of people.
- Technical Engineering Survey.
- Direct field observation of selected earthquake affected settlements.

Specific focus was given to perspectives of local stakeholders such as communities, municipal authorities, and real victims. This study mainly focuses on the urban areas of 14 severely affected districts by the Gorkha earthquake. The detailed research methodology adopted in this study are explained below.

Fig. 1.1 Research Methodology



2.1 Information Requirement Analysis

A requirement analysis was performed by the study team to identify data and information needed. Potential sources such as literature reviews, satellite images and analogue maps, key

informants interviews, focus group discussions, in-depth case studies, technical engineering surveys, field observation, and photographs were discussed.

2.1.1 Selection of Urban Settlements

A selection criteria of urban settlements was developed in consultation with HRRP/HPM to choose urban settlements in Kathmandu Valley, Charikot, Chautara, and Gorkha (Annex 2.1). Four settlements were selected from City Inner Core known as Kathmandu Valley Core (i.e., core areas of Kathmandu, Lalitpur, Bhaktapur and Thimi), one settlement from Peri-Urban (i.e., Gongabu), and four settlements from Historic Towns on Fringes (i.e., Panga, Khokana, Sankhu, and Bungamati) to assess the impact of the earthquake on heavily damaged urban settlements.

Additionally, a densely populated and haphazardly architected area, where houses were not severely damaged by the earthquake (i.e., Ghattekulo) was selected to assess the earthquake's impact. The main market area of Charikot, Chautara, and Gorkha were selected to assess the assessment of earthquake on urban settlements outside Kathmandu Valley. Two informal settlements from Kathmandu Valley and one informal settlement from Gorkha market town were selected to assess the impact of earthquake on informal settlements. Similarly, one apartment building was selected for FGDs to assess the earthquake impact on its residents. These settlements were grouped into seven sectors – KV core, Peri-urban, densely settled, historic town, market town, informal settlement, and apartment building.

2.1.2 Preparation of Base Map

High resolution satellite images integrated with of GIS-based data has been used to: delineate the study area; to identify the potential areas used for temporary shelters with time series images before and after earthquake; to estimate the area and houses included in the urban settlement; to use as a base map for field observation during technical engineering surveys; and to map the apartment buildings in Kathmandu Valley.

These base map was prepared using online mapping services systems such as Google Earth, Open Street as well as available GIS data on a municipality's ward boundaries. Profile maps of selected urban settlements are presented in Annex 1.

2.1.3 Clustering of Settlement

Data and information collected through the preliminary field visits and GIS maps were updated based on those data points. This led to the stratification of selected urban settlements into a number of clusters for focus group discussion (FGDs) for a detailed field survey. The stratification was mainly based on the major characteristics of urban settlements and affected communities such as displaced communities residing in temporary shelters, core settlements with severe damages, displaced local residents from core settlements, poor renters, marginalized communities, local residents residing in damaged, risk prone houses, local residents of heritage conservation areas, house pooling groups, etc. The number of clusters and their respective major characteristics is available in Annex 2.2.

2.1.4 Preparation of Checklist and Questionnaire

A checklist was prepared for the preliminary field visits in selected urban areas. It provides a list of available documents, names of support organizations, and challenges faced by

earthquake victims, etc. It also includes a guideline to mark the location of temporary shelter, name of important location in the map and list of important documents to be collected from the study areas (Annex 2.3).

After the preliminary visits at selected urban areas, the study team developed a semi-structured questionnaire for a detailed field survey through FGDs. The questionnaire seeks details on short-term and long-term impact of the earthquake on local residents based on issues and concerns (e.g., humanitarian and livelihood, challenges faced in temporary shelters, damage to culture and heritage, etc.). It also provides details on the type of recovery initiatives taken up by local stakeholders and support organizations post-earthquake and issues and challenges during reconstruction (Annex 2.4).

The study team also designed two types of checklists for the technical survey to understand the status of reconstruction at the selected urban settlements until the time of the survey. They are (i) the status of damaged but neither demolished nor restored buildings which have been labelled as not suitable for use, (ii) the status of buildings under construction post-earthquake (Annex 2.5).

2.2 Collection of Data and Information

The secondary data and information were collected from relevant stakeholders such as Nepal Reconstruction Authority, Housing Division of Department of Urban Development and Building Construction, UNHABITAT, municipalities, etc. The list of documents collected and reviewed is provided in the References. The primary data and information were collected from preliminary field visits, FGDs, KIIs, technical surveys, and direct field observation.

2.2.1 Preliminary Field Visit

The main objectives of the preliminary field visits were to introduce the objective of this research study to the respective municipalities, to collect secondary (quantitative and qualitative) data from local stakeholders such as municipalities, wards, other relevant government and non-government agencies, CBOs and Civil Societies; to have direct interaction with communities and earthquake victims; to identify particular locations of heavily damaged areas, temporary shelters, heritage sites, core settlements, informal settlements and marginalized communities, etc. The selection of key informants for the KIIs were made during the preliminary field visits to the selected urban settlements.

2.2.2 Detailed Field Survey

The main objectives of the detailed field survey were to collect primary data and information on the impact of earthquake, recovery initiatives taken, and status of housing reconstruction through KIIs, FGDs, and in-depth case studies. The data and information collected so far during the detailed field surveys is qualitative in nature.

Key Informant Interview

The key informants for interview were identified during preliminary field visit in the selected urban settlements and interviews were conducted during the detailed field survey. Majority of key informants were municipal authorities, ward secretaries, active community members, school teachers, local club members, local NGOs representatives, and volunteers involved in

rescue and relief of earthquake, etc. The KIIs were very informative and provided secondary information required for the study. The list of key informants is provided in Annex 2.6.

Focus Group Discussion

Focus Group Discussions (FGDs) were carried out by the study team at all major clusters within the selected urban settlements using semi-structured questionnaires. The FGDs focused primarily on the earthquake's impact on housing and livelihood of the selected communities, problems they have been facing, rescue and recovery initiatives taken, and major issues and challenges for reconstruction, etc. A total of 32 FGDs were conducted.

FGD participants were very cooperative and were frequently emotional to meet the study team for the first time on their ground to listen to their grief and to see how they were coping with hardships. They were very hopeful of getting a positive and prompt response from their interaction.

Case Studies

The study team interacted with several victims over a wide range of urban settlements in study area and prepared detailed case studies. The victims have shared their experiences from the impact following the immediate aftermath of the earthquake to the challenges they have faced since then. However, some of them were not comfortable in having their private grief published in a report due to social prestige. In such cases, only major issues and challenges faced by them were included in the report and few cases are briefly explained with their consent in Annex 2.7.

2.2.3 Technical Survey

A team of engineers visited urban settlements and assessed the status of damaged but not demolished houses and newly constructed buildings using the checklist and guidelines from field observations. A technical survey was carried out at all urban settlements except Ghattekulo, since there was minimal damage.

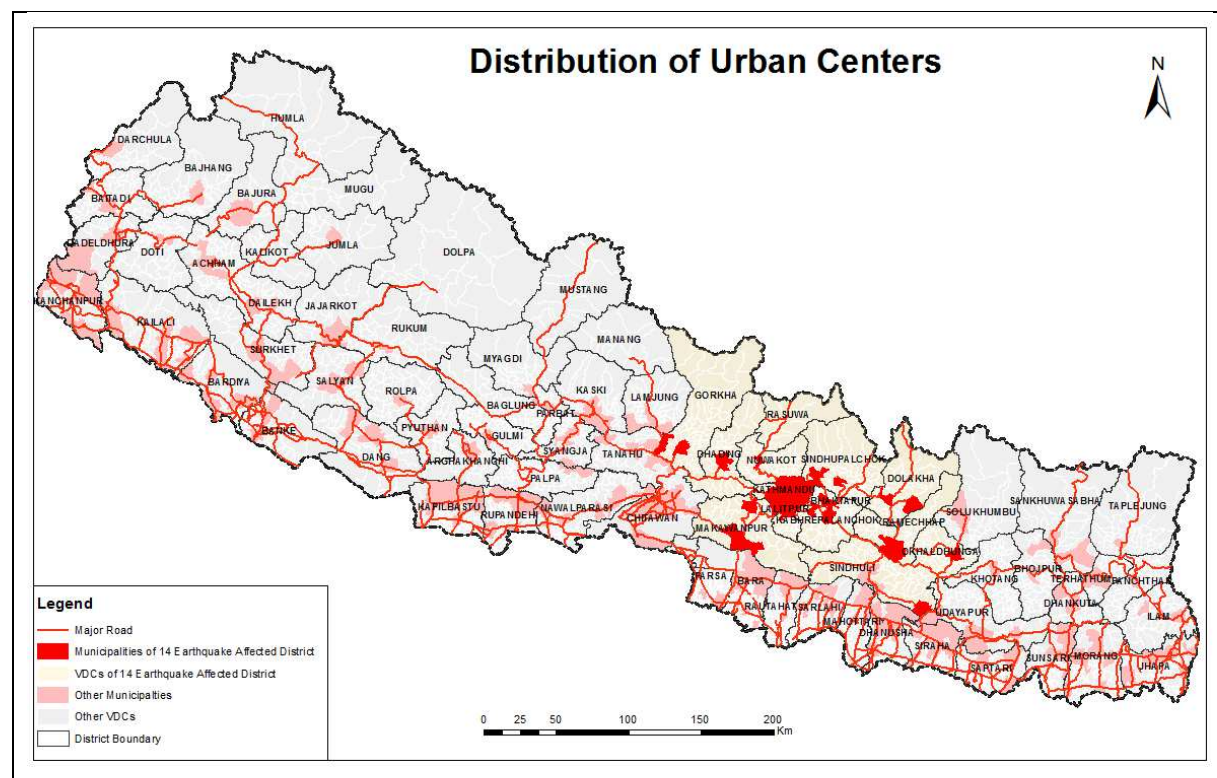
2.3 Analysis of Data and Information

The primary and secondary data and information, KIIs, FGDs, and technical surveys were collated, compiled and analysed for each individual urban settlement and separate urban sector. The land area and number of houses within each urban settlement was estimated by using GIS maps and ward level data. In some cases, the urban settlement included only part of a ward. In such case, the percentage of built up area comprising of building footprint was computed for the area inside the settlement and used to estimate the total number of houses in the settlement.

The quantitative and qualitative data of each urban settlement is presented in the respective profiles in Annex 1. The profile contains settlement area with percentage of municipal area covered, name of corresponding municipality, wards included (full or partly), total number of households with percentage of municipality, owned and rented households, male and female population, major ethnic communities, major economic activities, major construction types of houses, name of major tangible and intangible heritage/monuments, number of damaged houses (full and partial), number of human casualties and injured people. The photographs taken during the preliminary and detailed field visits, technical surveys were collected and have been documented.

3 Impact Assessment of Earthquake

It has already been mentioned earlier that the urban areas are functioning as the service centers for neighboring VDCs and small settlements in following map shows distribution of Urban Centers in Nepal. There are all together 217 urban areas declared as Municipalities, Sub-metropolitan cities and a Metropolitan City by the Government of Nepal till 2015.



Severely earthquake affected 14 districts alone have 41 urban areas or municipalities comprising 3 million people with 0.74 million households and urban population density of 1,382 per/SqKm. These 41 urban areas consist of 60% households with 57% population of the total earthquake damaged districts (Fig 3.1).

The data on damaged households of earthquake affected districts was collected at district and municipal level from corresponding municipalities, DDCs and DDRCs with the help of HRRP team. This reveals that approximately half million households were fully damaged, of which 33% damaged was in urban area (Annex 3.2). In this context it is imperative to assess the impact of earthquake in urban area separately.

The area of the new municipalities has been estimated by using GIS maps and the population and number of households were totalled from the known population figures of the erstwhile VDCs from which these municipalities were created based on the 2011 census (Annex 3.1).

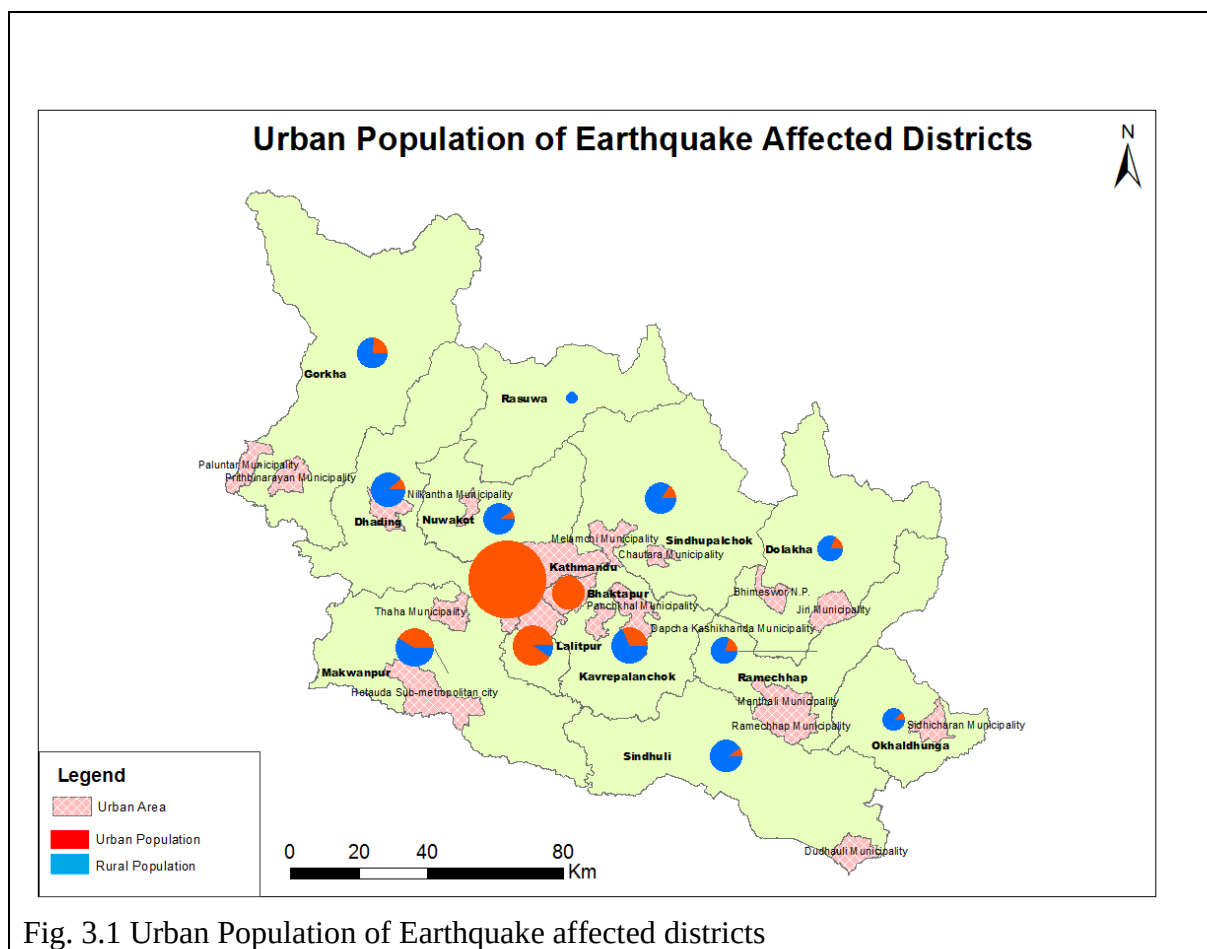


Fig. 3.1 Urban Population of Earthquake affected districts

The data on damaged households in earthquake affected districts was collected at both the district and municipal levels from respective municipalities, District Development Committee (DDC) and District Disaster Relief Committee (DDRC) with the help of HRRP team. The data reveals that approximately 500,000 households were fully damaged, of which approximately 33% were in urban areas (Annex 3.2). There are different causes of building damages. This study has compiled a set of photographs of damaged building with the causes of damage which is presented in Annex 3.3.

This chapter explains the impact of earthquake on each of the seven sectors separately based on KIIs and FGDs carried out at the 31 clusters within the 16 urban settlements. The basic information and urban settlement profile is presented in Annex 1 with References. The seven sectors and their corresponding urban settlements with number of clusters for FGDs are as follows:

1. KV Core settlement - *Kathmandu Core (5), Bhaktapur Core (3), Lalitpur Core (3), Thimi Core (1)*
2. KV peri-urban settlement – *Gongabu (1)*
3. KV densely settled area – *Ghattekulo (1)*
4. KV Historical towns in fringe – *Panga (2), Khokana (2), Sankhu (2), Bungamati (2)*
5. Market town out of KV – *Charikot (2), Chautara (2), Gorkha (2),*
6. Informal Settlements – *Bansighat (1), Khadipakha (1)*
7. Apartment Buildings - *Sunrise Apartment (1)*

The seven sectors comprises of more than 80 thousand household with 55% rented and 45% owned households, population of more than three hundred thousand. The Number of damaged building in these sectors is about thirty eight thousand, of which 56% buildings were fully damaged and 44% were partially damaged (Table 3.1). The information on Area, Population, Household and Damaged or corresponding Municipalities of Selected Urban Settlement is shown in Annex 3.4.

Table 3.1 Households, Population and Building Damaged of Seven Sectors of Study Area						
	KV Core	Peri-urban	Densely Settled	Historical Town	Market Town	Grand Total
Total Households	61039	3684	7438	4872	3620	80653
Owned	25543	1354	2408	3222	2421	34948
Rented	34204	2277	4874	1599	1105	44059
Population	258692	14081	26653	19095	12634	331155
Male	134696	7348	14500	10100	5998	172642
Female	123995	6733	12153	8995	6636	158512
Building Damage	27843	1363	25	5063	3793	38087
Full	15120	150	25	3724	2360	21379
Partial	12723	1213		1339	1433	16708

3.1 Kathmandu Valley Core Settlement

Kathmandu Valley core settlement represents the city's inner core area which has major tangible heritage and has an intrinsic intangible cultural significance. There is a mix of old market centres, traditional private homes and public houses, symbols of religious history, and is the cradle where culturally valued ceremonies are performed daily. This includes mainly old settlements of Kathmandu, Patan, Bhaktapur, Thimi, and Kirtipur having history of origin dated back to pre Lichchabi period.

Early human settlements (*grama*) located on fallow high ground during the 'Kirata'¹ period (pre-historic time) over time evolved into commercial centres (*dranga*) during the 'Lichchhavi' era. They were further extended, connected by roads and fortified by gates and walls during the 'Malla' era, which led to the formation of medieval town centres in Kathmandu Valley (Regmi, 1965; Oldfield, 1974; Wolfgang, 1976; Malla, 1978; Slusser, 1982).

All this coexisted in harmonious balance between man-made structures and the surrounding natural ecology. The integration of open and built spaces, along with the architectural composition of buildings and monuments have not only supported the existence of a culturally advanced urban society of 'Newars' but also have proved that those builders were extraordinarily skilled at integrating visual and functional principles with societal needs (Hosken, 1974).

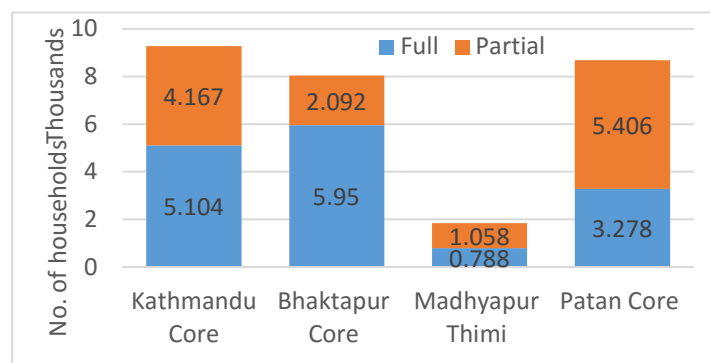
For this study, Hanumandhoka Durbar Square, Mangal Bazaar periphery including Patan Durbar Square, Bhaktapur Durbar Square, and Balkumari Thimi core areas were selected from the set of Kathmandu Valley core settlements. These settlements comprise mainly of Newar communities since their inception. At present, the major economic activities in these settlements are private businesses, services industry, handicrafts, skill based industries, rental units, agriculture etc. Twelve FGDs were carried out to assess the earthquake's impact on this sector and the following sections are based on those discussions. The following areas have been researched: humanitarian and livelihood; public infrastructure; culture and heritage; marginalized group and poor renters.

3.1.1 Housing and Livelihood

Building Damage

The number of households damaged was comparatively high in the core areas of Kathmandu Valley, which consists of compact, dense and traditionally built houses. More than 27 thousand households were affected, of which, 54% were fully damaged in core areas (Fig 3.2). Consequently, the death toll was also very high. During the FGDs it was apparent that within

Figure 3.2 Number of Building Damaged by EQ in KV Core Settlements



¹ The political historic periods of Kathmandu Valley (and Nepal): Kirata period (prehistoric), Lichchhavi period (1st - 9th century), Malla period (13th - 18th century) and Shah period (18th century - 2007 and the Rana ruled between 1846 to 1951).

each community, residents lost their family members and many were injured. The collapse of the ‘Kasthamandap’ took lives of 12 persons in addition to 37 people in KMC Ward 20.

Many Unreinforced Masonry Buildings located in the core areas of all the four settlements were badly affected by the ‘Gorkha’ earthquake. Typical forms of damage to such structures are of the following types.

- Many houses built with bricks (mud or sun dried) in mud mortar either completely collapsed or were seriously damaged. Houses also collapsed from lack of timely maintenance and repair. Several traditional houses had undergone various stages of transformation from vertical division of original houses among brothers and haphazard renovation and extensions.
- In the studied settlements, failure of upper floors of the traditional houses was also observed due to excessive load on the upper floors (mainly from reinforced cement concrete floors) to meet increasing space need.
- Collapse of side walls (full or partial) in the houses built with masonry and load bearing wall system was also noticed. The traditional houses in the valley built during Malla and Rana periods have load bearing walls at two sides only. Hence, it is possible to divide them vertically into different parts.
- Lack of corner ties and weak masonry bonding at the corner walls has led to development of vertical cracks on the walls along the joints in many traditional houses in the core areas.
- Many old houses developed diagonal cracks across the masonry walls due to inability of the structures to resist the tensile stresses developed due to combination of horizontal and vertical loads.
- Many old houses were deformed in the direction where no building existed. Wooden and bamboo flooring with mud mortar are poor in horizontal stiffness.
- Peeling of the brick from the load bearing masonry walls and part of the wall coming down without complete destruction of the house were also observed

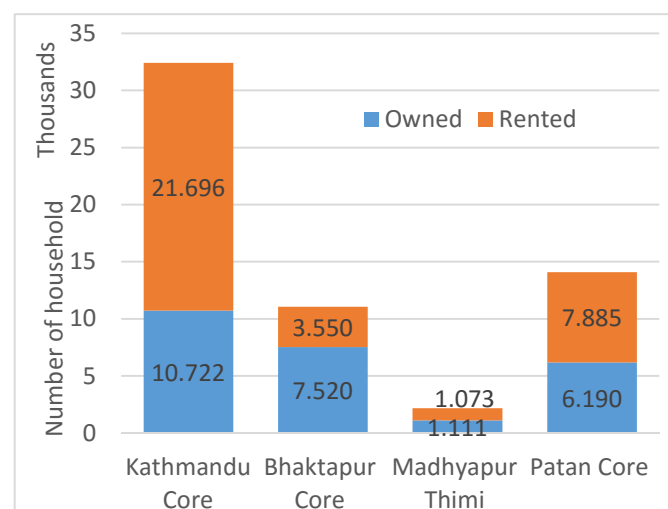
Temporary Shelter

A majority of earthquake affected residents, either house owner or renter, spent many nights outside their houses in its immediate aftermath. The core settlements comprised of more rented households (about 56%) than owned households (46%) (Fig 3.3).

Our team discovered that since then (i.e., ten months post-earthquake), most victims have been found in temporary shelter, which can be categorized in the following ways:

- Majority of earthquake victims are still living in temporary shelters constructed by

Figure 3.3 Number of owned and rented households in KV core



corrugated galvanized iron (CGI) sheets, bamboos, tents, etc. in private land mainly in Bhaktapur. These sheds have basic facilities such as a common toilet and water tanks.

- Some earthquake victims have taken shelter with their relatives at nearby locations by either renting or free of cost.
- Poor renters in Kathmandu continue to reside in partially damaged old houses, which are supported by temporary wooden logs.
- Those residents who could not afford to pay rent were forced to return to their damaged houses. Many victims in all four municipalities belong to this group.

Livelihood

People living in the core settlement areas of Kathmandu Valley have different professions, which include services, trade, handicraft, rent, etc., and these vary from one cluster to the next. Loss of income source leading to economic hardships for an individual household and family has long term serious effects. Some major impacts are categorized as below:

- Those engaged in small businesses and handicrafts or masonry and carpentry lost their jobs due to the economic slowdown, distinct reductions in tourism flow and construction activities.
- Agriculture based communities could not work in fields as they were covered with temporary shelters.
- Many small businesses that were located on the ground floor of buildings had to be shut down due to collapse or partial damage of the houses.
- Many households mainly based on rental income lost their income as well as the house.
- Poor renters and others relying on daily wages also lost their income due to lack of work.
- Poor renters in Kathmandu have suffered terribly due to loss of business in one hand, and on the other hand, they need to pay higher rent.
- In a situation of loss of household income or other impacts to livelihood, earthquake victims needed to find additional money either for building temporary shelters or renting alternative living accommodation. Significant amounts of money were also spent on placing temporary supports to prop up damaged houses as well as to demolish the upper RCC floors, particularly in Bhaktapur. Most of them received partial support from GoN (NRs. 15,000) for buying CGI sheets and/or non-government organizations for constructing temporary shelters.
- A majority of households living in the core areas of the Kathmandu valley have taken small loans from their relatives, community cooperatives or women's groups for buying food provisions, paying school fees or on renting space to store their belongings. They have so far managed to pay back the interest only.
- Due to the displacement of people, the volume of business has reduced.
- Many fear an impending food shortage due to lack of agricultural work and failure of rains. Due to lack of funds, purchasing food items will be challenging.
- Lack of privacy, hot interior space during summer and cold in winter, noise while raining due to CGI sheet covering are some inconveniences caused. In some cases, children had to drop out from school due to loss of family income.
- Women during menstruation time suffer a lot not only due to lack of privacy in the shelter but also lack of water and other facilities in the common toilet.
- Loss of income, and confusion about government's reconstruction policy have created mental stress in elders.

3.1.2 Public Infrastructure

There has been no major infrastructure damage, except for a few broken water supply pipes, telephone lines and electricity cables. Public facilities, i.e. public transportation, electricity supply and private sector water delivery (tankers) were hampered for many days. Schools were closed for many weeks. However, neighbourhood retail shops opened almost immediately and the mobile communication network was constantly.

3.1.3 Culture and Heritage

Culture has played a key role of in architecture and growth process of Kathmandu Valley. 'Lichchhavi' towns developed at high ground were compact and dense. There exists an inventory of 1200 traditional houses with heritage value in Kathmandu according to Inventory and documentation of private heritage homes and public monument of Kathmandu (KMC, 2015).

The implication of earthquake on culture and heritage is many folds.

1. Many World Heritage Sites and the surrounding ancient monuments either collapsed or were badly damaged. The affected monuments in Kathmandu Durbar Square include Kasthamandap, Panchtale temple, the upper levels of the nine-storey Basantapur Durbar, the Dasa Avtar temple and two dewals located behind the Shiva Parvati temple. Some other monuments, including the Kumari temple and the Taleju Bhawani temple were partially damaged. Similarly, the top of the Jaya Bageshwari temple in Gaushala and some parts of the Pashupatinath temple, Swyambhunath, Boudhanath Stupa, Ratna Mandir inside Rani Pokhari were also destroyed.

In Lalitpur, the Char Narayan Mandir, the statue of Yog Narendra Malla, a pati inside Patan Durbar Square, the Taleju temple, the Hari Shankar, Uma Maheshwar temple and the Machhindranath temple in Bungamti were destroyed.

In Bhaktapur, the Fasi Deva temple, the Chardham temple and the 17th century Vatsala Durga temple, were damaged. Weak structural material, i.e., brick in mud mortar, use of wooden posts and beams coupled without tie beams on the perimeter of the buildings at different levels including lack of maintenance are the major reasons behind such damages.

2. Many national and local festivals were short-cut and witnessed dampened enthusiasm. Traditional 'jattras' or processions did not attract large crowds. It was difficult to raise funds with donations and the fear of another earthquake was all pervasive.
3. Many family rituals have been shortened. Important festivals of Dashain and Deepawali were celebrated just to continue the process.

3.1.4 Marginalised Group

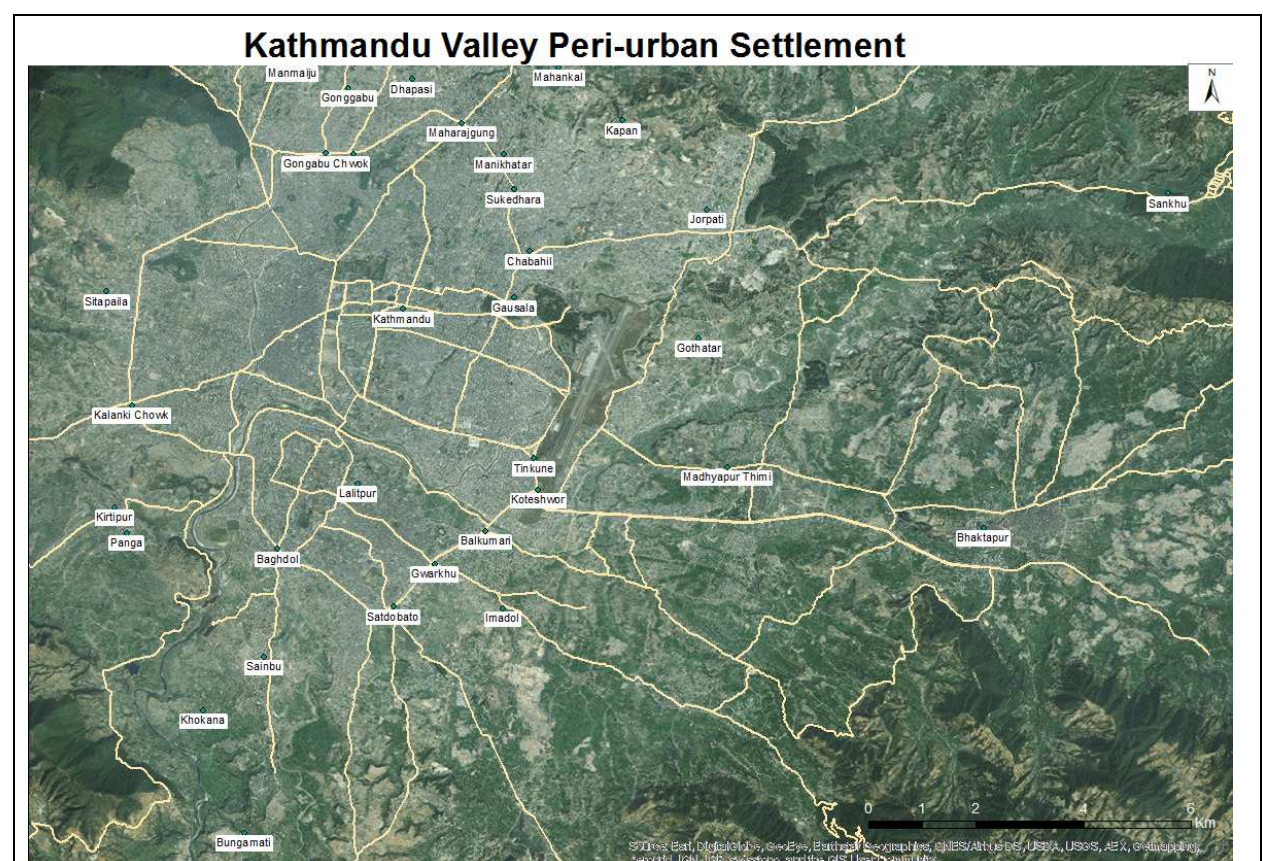
Though marginalised groups of people have not been distinctly noted in the studied communities, they do exist in isolation. However, they were not discriminated against either during the rescue and relief operations or during their stay at temporary shelters. They are well treated by other community members.

3.2 Kathmandu Valley Peri-urban Settlement

The growth of peri-urban settlements in Kathmandu dates back to the late 1970s and early 1980s when Kathmandu experienced a large impetus in growth from increased concentration of government activities, growth of tourism trade and industrial development, i.e., carpet industries. The 27 km Ring Road had been completed and some of the "peri-urban" settlements grew at nodal points where radial roads diverging from central Kathmandu intersected with the Ring Road. The growth of Koteswor, Satdobato, Gongabu, Dhapasi, Mandikatar, Suke Dhara, Jorpati, Sinamangal and other areas can be attributed to this urbanisation trend.

The peri-urban settlements were previously outlying settlements of the central city areas of Kathmandu and Lalitpur. Their buildings are newer on plot sizes which on average are larger than those in city core areas. Infrastructure services, e.g., water supply, sewerage and drains arrived after the physical township developed, rather than preceding it as it is common in Kathmandu and other unplanned urban areas.

Peri-urban buildings are characterised by modern RCC frame structures followed by load bearing walls. However, despite modern construction techniques, there was extensive damage to the buildings. In this study, Gongabu has been selected for assessment of earthquake impact as this settlement is different from other settlements because of earthquake intensity, its geological formation and commercial nature.



3.2.1 Housing and Livelihood

Building Damage

Building damage in the peri-urban areas has been disproportionately high notwithstanding that these are recently developed areas with relatively newer buildings. In Gongabu, a total of 1,363 buildings were affected, of which 150 were completely damaged. Currently 13 undemolished damaged buildings are still standing of which five are being partially used by the owners, one by renters and the remaining seven are not being used for any purpose. For safety, two buildings have been partially dismantled and three are going to be reconstructed. Of the remaining three buildings, two are planning for partial reconstruction and one has not initiated any safety measure. One of participants of FGD in Gongabu, has lost seven family members of the total 14 due to complete collapse of his modern five and half storey RCC frame structure building from the earthquake.

Temporary Shelter

As peri-urban area is newly developed, most of the building types are RCC frame structure or load bearing wall system. In Gongabu, the share of RCC frame system houses accounts for 60% followed by wall system accounting for 30%. The remaining 10% is old traditional type of housing mostly built on mud mortar. Currently most of the affected families have settled themselves either in rented space or in relative's houses. There were no temporary shelter within the selected settlement of Gongabu at the time of field survey.

Livelihood

The major source of livelihood is service at 70% followed by trade and commerce at 25% and agriculture including daily wage labour at 5%. The earthquake has greatly affected local businesses and enterprises. Poultry farming entrepreneurs had to sell their chickens at very low prices due to unavailability of feed. A large number of chickens - 12,000 to 13,000 - also died during the earthquake. House owners who had previously rented out their houses did not get tenants for about 5 months. Hardships during the earthquake have been experienced by all relating to food, shelter, medical treatment, etc., in particular, by the poor since they generally do not stock such items for an impending emergency.

3.2.2 Public Infrastructure

Public infrastructures and services in peri-urban areas are relatively advanced. The infrastructure can be upgraded or repaired with greater ease than in central core areas. The households are connected to basic infrastructure services, i.e., electricity, drinking water supply, black topped roads, etc. There were no breakage of water supply, drainage and no breaking of electric poles. Only electricity supply was temporarily interrupted during the earthquake.

3.2.3 Culture and Heritage

Some of the significant monuments and historical sites in the area are Mhyapi, Nanak math, Kapurdhara (stone spout), Natyeswor. Balaju Garden, Manmai ju, Purano Guhyeswori, Tokha and Budanilkantha. Earthquake has not affected except a crack in Kapurdhara.

3.3 Kathmandu Valley Densely Settled Area

In the decade from 1970 - 1980, Kathmandu saw an increase in employment opportunities with additional expansion of government machinery, development of trade and tourism, and the plethora of carpet industries. Now, with further growth in urban activities and concentration of central business development activities along Kathmandu's west-east axis and south-north axis, these previously sparsely developed pockets evolved into densely populated zones with higher building bulk and started to accommodate more people. This development is characterised by modern mid-rise RCC structures on narrow streets with very little or no open space.

While Gorkha earthquakes had relatively moderate little impact on such areas in terms of building damage and death toll, however it led to a general sense of public awareness in terms of vulnerability and disasters. The assessment of impact of earthquake on such settlements might be useful for planning disaster management response in the future for densely settled areas of Kathmandu Valley. Thus, Ghattekolu (KMC Ward 32), which is close to the city core was selected. It was a sparsely developed area squeezed in between Dillibazaar - Maiti Devi and Singha Durbar's north-eastern corner. However, with the opening up of the Anamnagar access from Hanumanthan (south-eastern corner of Singha Durbar) to Ghattekolu's mouth on the Singha Durbar side, the area experienced rapid growth and transformation.

3.3.1 Housing and Livelihood

Building Damage

Fortunately, Ghattekulo suffered relatively little damage during the earthquake. One residential structure was affected, but it did not fall down. Afterwards, the owner demolished it himself. Major impact was that it brought daily activities to a near standstill. Many residents are immigrants from other parts of Nepal. During the initial days following the earthquake, there were discussions of demolishing storeys and/or parts of buildings that were not compliant with the National Building Code. A group had even considered to move jointly to a location far from the city's dense area. But later all these initial "commitments" were forgotten. But the feeling that safety is the primary concern is still strong among residents.

Temporary Shelter

During the immediate aftermath of the earthquake, many residents sought refuge in Tundikhel (about 1.6 km west), Bhrikutimandap, and Singha Durbar due to lack of open spaces nearby. Some residents went to stay with friends and relatives in comparatively safer environs. Initially people spent majority of their time in these temporary shelters or accommodation away from their own houses. Some had even camped out on available private grounds under tarpaulins. With the passage of time, the time interval between successive aftershocks increased and their intensity abated. Soon after, families started returning to their houses in Ghattekulo on a trial basis. During these trips back they would either change their clothes or perform other necessary chores to keep their sanity or semblance of a normal life. However, they were concerned that their neighbourhood lacked open spaces. So, there was no place to run to during a major disaster as mentioned in FGD.

3.3.2 Culture and Heritage

A few Ganesh shrines and other small temples are present in these areas, but do not carry much significance as newly built up area. This settlement has mixed culture.

3.4 Kathmandu Valley Historical Towns in Fringe

Kathmandu Valley comprises of many historical towns which are located mostly along the outer rim or fringe of the city. Historically, they were self-sustained satellite towns² part of a complex urban corridor system of the erstwhile kingdoms of Kathmandu Valley. These towns are also under transformation in context of building type. New RCC frame structures are coming up in the town peripheral areas, whereas in the town centre, the tiled roofs and CGI roofs of many traditional old houses are being replaced by concrete slabs and vertical division of old houses while transferring parental property to children. Some of these historically important towns are Bungamati, Khokana, Pharping, Chapagaun, Thecho, Thasi, Thali, Harisidhi, Lubhu, Tokha, Sankhu, Changunarayan, Panga, etc. This study has selected four historical towns – Bungamati, Khokana, Sankhu and Panga for assessment of Gorkha Earthquake in these settlements.

This study conducted KIIs and FGDs at selected urban settlements within these historical towns. These ancient settlements are laden with socio-cultural and religious history. Many have existed since the Lichchavi period (400 - 750 CE) and mainly comprise of predominantly residing Newar communities. The majority of houses were constructed of mud, mortar and wooden crafted doors and windows. These construction practices predate modern architectural styles and material. They consist of archetypal Newari architecture from the Malla period (1201 - 1769 CE) and have high heritage significance and cultural value. These settlements consist of a number of culturally, artistically, and historically significant buildings.

Historical towns are very popular with tourists for their indigenous architecture and design. Majority of residents have inherited a traditional handicraft skill from their ancestors. These settlements not only serve as a major tourist attraction but also play key role in promoting the national tourism industry. For example, in Bungamati, which is well known for its traditionally inherited wood carving skill, majority of households are engaged in the wood carving business besides farming. The residents of Bungamati are proud of their workmanship and ability to earn their livelihood within their community. They are also thankful they do not have to immigrate to foreign lands or leave their settlements in search of work. Similarly Khokana is popular for its traditional mustard oil processing. There are many historical towns in the fringes, each of which are historically noteworthy. Their cultural and religious value make them ideal tourist zones as well and provide for an additional income stream for local residents.

3.4.1 Housing and Livelihood

Building Damage

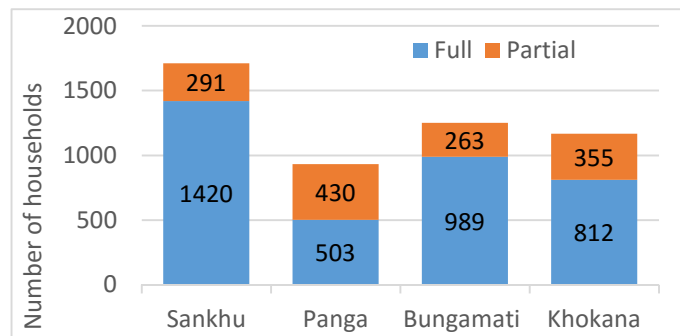
Historical towns of Kathmandu Valley were severely damaged by the earthquake. About 5063 buildings were damaged, of which 74% were fully and 26% were partially damaged in these settlements (Fig 3.4). Such houses mainly built with mud bricks or sun dried bricks in mud mortar almost completely collapsed down to foundation. Even those which remain standing are seriously damaged and have been rendered unsuitable for habitat. Some traditional houses, which had an additional floor constructed of cement mortar experienced only their top floor collapsing due to excessive load on the upper floors. There were also cases of collapsed or cracked side walls in houses built with masonry and a load bearing wall system.

² A **satellite town** is a concept in urban planning that refers essentially to smaller city areas which are located somewhat near to, but are mostly independent of larger metropolitan areas.

In Panga, a few of the newer buildings based on RCC structures also collapsed because of design failures. During KIIs and FGDs in selected historical urban settlements, the team was informed that approximately 51 people died in four settlements and many were injured. However, in Khokana, FGD participants claimed that 35 people died after EQ due to heart disease, pneumonia and other accidents, i.e. tumbling on the debris, etc.

However, this information has not been recorded either at the ward office or the municipal offices.

Figure 3.4 Number of Buuilding Damaged by EQ in historic towns



Temporary Shelter

In the immediate aftermath of the earthquake, all residents remained outdoors. Some camped under temporary tarpaulin shelters on surrounding farm lands, while others on open public grounds nearby their residential settlements for many days and nights at end. Only after a few weeks, a handful of NGOs and INGOs started to support the construction of temporary shelters of bamboo, plywood, CGI, etc. However, some victims were not able to access such temporary shelters and were forced to stay under tarpaulins. The temporary shelter was limited to one room per household. Families were able to cope for the first few months, but was not possible for an extended period of time, especially for a large family. Eventually, residents started to use their damaged houses during the day hours. They started staying even during the night after a few months despite being vulnerable to the continuing aftershocks. This scenario repeated itself at each selected historic town settlement.

Livelihood

Most historical towns along the fringes are reputed for their traditional inherited skills of metalwork, wood crafting, stone carving and processing of typical local food items (e.g., mustard oil). While farming and agriculture provide the major income source for these settlements, the handicraft business is also a steady income stream.

The earthquake affected not only their daily business but also their agriculture production was placed in abeyance, since they could not farm due to the land being used for providing temporary shelters. In Bungamati settlement, almost every house had a small wooden handicraft (known as *gyasa*) workshop on the ground floor. Many residents had these workshops damaged and their tools were buried under the debris. Although they could retrieve the tools after a few months, they did not have the space to continue their work, since it is not possible to work in temporary shelter. In almost every historic town settlement, the tourism industry has been severely affected. During the FGD sessions, local residents expressed their grief of loss of income together with houses.

Many injured victims are still under long term treatment and suffer due to weakness and other ailments. During the dry winter months, many children and the elderly living in temporary CGI shelters suffered from pneumonia and cold fever. About 30 people lost their lives during the post-earthquake phase in Khokana as mentioned in FGD. On the other hand, some children were forced to discontinue the school due to loss of parents' incomes. This was especially true

for children who were seeking admission to college right after their class 12. Instead they entered the job market in hope of being able to provide support to their family.

During the immediate post EQ time frame, temporary shelters provided great support to the victims. Such a shelter has an external common toilet and lacks potable water. However, after a few months, the victims experienced inconveniences and felt unsafe in such temporary shelters. In some shelters, there was no electricity, which prevented residents to exit after evening hours. Women suffered lack of privacy, water and other facilities in the common toilet, especially during menstruation. Instead, they were forced to use their damaged houses which were still vulnerable to minor aftershocks. The stress of reconstructing the damaged houses led the depression to many elderly males and get drunk according to female family members in these settlements.

3.4.2 Public Infrastructure

There have been no major infrastructure damages as compared to private citizen houses in these settlements. Few mud mortar based school buildings were had some cracked walls. These facilities have been transferred to temporary sheds. In Bungamati, a water pipeline was damaged, however the community repaired it themselves.

3.4.3 Culture and Heritage

The historical towns along the fringes of KV are culturally significant. They have great social and heritage value. The April EQ damaged many monuments (e.g., temples, *pati*, *dhunge dhara*, stupa, etc.) within the settlements. Beside these monuments, many historical artistic private buildings located near heritage sites were either partially or fully damaged.

During FGDs with residents neighbouring heritage sites, they expressed their worries of the possibility of the complete loss of ancient building architecture, especially if during the reconstruction process, a modern building style is adopted. According to them, the officially listed monuments as heritage will eventually be maintained by the government. However, the retention of a private building's of architecture with priceless heritage value will prove to be the greater challenge. Due to the high cost of building such houses, owners may wish to build cheaper commercial buildings instead of maintaining the same architecture. This will have great impact on the livelihood of heritage sites.

3.4.4 Marginalised Group

The effect of the EQ on marginalized communities was also studied. They too were equally affected by the April EQ like all other local residents of their respective settlements. During KIIs, we discovered that there has been no discrimination among the victims during the rescue and recovery process. The marginalized populations were treated on par with all other affected residents. They also resided in temporary shelters in close proximity to their settlements.

3.5 Market Towns outside Kathmandu Valley

Nepal's urban system is dominated by the capital region of Kathmandu Valley, where majority of the nation's urban dwellers are located. After Kathmandu, the next order of urban towns lie in the Terai region with its large number of urban dwellers. None of these market towns outside KV encountered any significant impact of the earthquake. The next order of towns are the hill towns. In the 14 districts affected by the EQ, there are 19 market towns. Some of them are: Baharabise, Chautara, Charikot, Dhulikhel, Gorkha, Manthali, Okhaldhunga, Panauti and Panchkhal. Many of these towns are market centres or developed into market centres since they were designated as district headquarters or some other significance was attached to them. It is also true that some of the market towns were higher order towns either historically or for other reasons. For example, Dhulikhel in Kavrepalanchok was Purba No. 1 during the Rana regime, and prior to that Gorkha in western Nepal was the capital of the Gorkha kingdom.

Three market towns – Gorkha bazaar, Chautara bazaar, and Charikot bazaar were selected to assess the impact of the earthquake on urban areas outside of Kathmandu Valley. The Gorkha municipality has developed along a linear pattern. Most of the houses have been built in the last few decades. The bazaar is on a gentle slope rising from Haramtari up to the Bus Park (Ward 2, 3 and 6). Since there already is a dense settlement in this area, Ward 8 is planned for the future expansion. In fact, this bazaar area is a recently developed area, and therefore there are neither old nor traditional types of houses as opposed to the old bazaar area. The major occupation of the population is trade and commerce (60%) followed by agriculture (20%), service (10%) and others (10%) including daily wage labour.

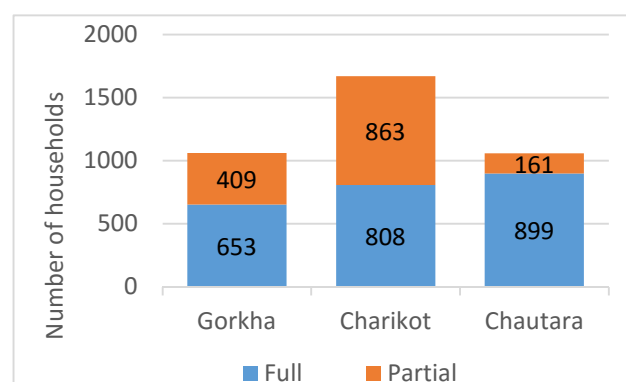
The bazaar area of Chautara Municipality is approximately along a SE to NW running ridge. As one approaches Chautara Bazaar from the Kodari Highway, the bus park is at the beginning of the bazaar. The Tundikhel is at the other end of the ridge-top. The distance from the bus park to the Tundikhel area is about 1.5 km. A few public government offices and the District Hospital are located around the Tundikhel area. Many relief agencies and other donor agencies have used or are still using this area as their staging area for their offices and other temporary facilities. The UNICEF office is located there as are some other agencies. The Charikot Bazaar is situated along the steep main road leading to Jiri. This area was newly expanded with RCC structure buildings.

3.5.1 Housing and Livelihood

Building Damage

Housing conditions in the bazaar area of Gorkha Municipality are relatively good in terms of their physical condition as most of the houses are new. The share of RCC frame structure and load bearing wall structure is 80% and 20% respectively in the urban area. The damages caused mostly to load bearing wall type of the houses mainly located in the old bazaar area on the way to Mathillo Durbar. In selected market towns, about 3793 buildings were found damaged, of which 62% were fully damaged (Fig 3.5)

Figure 3.5 Number of Buuilding Damaged by EQ in Market Towns



In case of Chautara, there was extensive damage of buildings mainly on second largest earthquake on 25th April with epicentre of Dolakha. It was noted that about 1060 private houses were damaged in Chautara bazaar alone. In Charikot bazaar, the earthquake has affected 1671 buildings, most of them were newly built RCC frame houses in this area.

Temporary Shelter

Immediately after the earthquake, open spaces in the bazaar areas of the market towns were used to set up temporary shelters, such as the Tundikhel areas in Chautara and Charikot, and open spaces in Gorkha. People also camped out in their own localities or *toles* during the days immediately following the earthquake. Open spaces within school grounds were also used. However, after the first few days as fear receded, most people left the temporary shelter camps and returned to their homes. Many built shelters from CGI sheets on their own land or adjoining properties. Many experienced difficulty in getting sufficient water to meet their daily needs for cooking and ablutions.

Livelihood

One of the immediate impacts of the earthquake was on livelihood. Any market town's economic contribution to trade and commerce is substantial. Ergo, the earthquake had a calamitous effect livelihood. Although the visible physical damage and other impacts were less in Gorkha and Charikot, the immediate impact was that it brought almost all economic activity to a near standstill. It took some time for business activities to return to normal. According to KIIs and FGDs in Charikot, where almost 50% of households are reliant on commercial activities for their livelihood, there was significant loss of income as the hotels and shops in the bazaar were shut down for 3 - 4 months.

In Chautara, where the physical damage was more intense, the destruction caused by the earthquake changed the basis for commercial life on the ridge-top bazaar. The central location of commercial activity was destroyed. Even if it were not destroyed, because of the uncertainty, it would have been difficult to conduct any commerce and trade. Later there were landslides in the surrounding areas and the border crossing at Kodari also shut down because of the damage on the Chinese side further adding to Chautara's grief. For the local farmers, who were not reliant on bazaar based commercial activities, it was almost impossible for them to return to their fields.

There has been a great impact from the influx of relief agencies into Sindhupalchok in the form of employment for locals as temporary labour, enumeration work or other employment in NGOs and INGOs.

Immediately after the earthquake, food supply was limited. There instances of food supply and stocks being lost in the earthquake rubble. A few days later, people were able to buy food from stores, etc. Some food relief was also provided through various agencies including the Municipality.

Damage to local hospitals and health centres affected the delivery of health services to the victims. The district hospital of Chautara provided services to patients under temporary tents since the main building was damaged.

The academic school year has been thrown into total disarray. Because of the damage to school buildings and other educational facilities, long term negative effects on education are expected. In Chautara, the Ratna Ganga School is running in temporary facilities. Other facilities affected are Bandevi School and 2 or 3 boarding schools. Normal repair has been carried out at Chautara

Bahumukhi Campus and will restart very soon. In Gorkha, damage to two school buildings interrupted children's studies for an extended period of time.

The most severely affected groups have invariably been children. Because they have never experienced anything similar, trauma effects on them have been much more pronounced. Children fear the repeat of the earthquakes and all those events which ensued after that. Apart from fear, other problems are minor headaches and a general feeling of lethargy. There were also many complaints of sleeplessness or waking up early in the morning and not being able to go back to sleep.

3.5.2 Public Infrastructure

The impact on public infrastructure in the market towns have not been severe. There was a minor impact on the water supply. In Gorkha, the water supply situation was further aggravated by the earthquake including damage of the main water source (*paniko mool*). In Chautara, the water source at Holchhe - Thal-kharka (under Jugal Himal) was affected. However, water service was restored within 4 days. Immediate temporary measures was to haul water from Hudrung waterfall (*jharana*), a nearby source. The Red Cross also assisted in this activity.

3.5.3 Culture and Heritage

Three temple complexes in the Chautara bazaar area were damaged. In the Bhimsen temple complex in the old bazaar area, two temples suffered minor damage. However, the adjoining pati (for storing temple and deity accessories and singing bhajans) has been destroyed because the structural supports upholding the pati were damaged. The Bandevi temple is located at the upper (northern) end of the bazaar. The pati adjoining the temple is vulnerable because the supports, which hold it up have weakened. The extensive damage to these temples has affected the jatras associated with them. The main jatras connected with the Bhimsen temple are during Krishnastami and Gai Jatra. Kartik is celebrated at the Bandevi temple. In Gorkha, Tallkot Kalika temple, Gorakhnath, Sri Vidhya temple, Pandit Pati, and Damai Pati were destroyed. In Charikot, an ancient gumba, built during the Dolakha Reign, dating back to 1618 B.S. has been completely destroyed.

3.5.4 Marginalised Group

In the urban areas of Gorkha, there are no informal settlers who have migrated from other districts as in the case of other major cities in Nepal. However, there are approximately 76 household/families staying in 17 sheds near the Bus Park who were displaced due to their land being acquired by Archaeology Department, GoN in BS2038/39. Now Prithivi Garden has been built on that acquired land which is located near Tallo Durbar. These households are all local people belonging to Dalit ethnic groups like Shreemal (Sarki) and Biswokarma. In fact all these families were given reasonable amount of compensation of NRs. 8000-9000 by the GoN during the acquisition as per decision of land acquisition committee. Unfortunately, being illiterate, none of these families bought land with the compensation and most probably spent all money on daily expenses.

The main source of livelihood of these families is daily wage labour followed by pig rearing, working at seasonal fruit stores, being a helper on local buses, etc. All these families have at least one/two pigs, whose sale supplements their family income. There are hardly 2/3 families with 12 to 15 members otherwise the average family size is 4 to 5. They fetch water from nearby source and toilet is constructed in all 17 sheds. However, proper water supply and adequate sanitation are the major problems in these informal settlements.

Post-earthquake all these families lost their source of daily wage labour for about 4-5 months which was the hardest part of their life. The earthquake caused partial and/or full damage to their sheds. However, with exception of a few, most of them have managed to repair their sheds. The commission formed to resolve the problem of informal settlers in the recent past had thoroughly surveyed their areas and had started the process of legalizing their status and distributing land certificates called 'Lal Purja'. But the entire process came to full halt for the reason unknown to them.

3.6 Informal Settlement

An Informal Settlement is defined as residential area where inhabitants have no security of tenure vis-à-vis the land or dwellings they inhabit, with modalities ranging from squatting to informal rental housing. The neighbourhoods usually lack, or are cut off from, basic services and city infrastructure. The housing units may not comply with current planning and building regulations, and are often situated in geographically and environmentally hazardous areas (UNHABITA, 2012). In order to assess the impact of Gorkha earthquake on informal settlers, two informal settlements of Kathmandu - Banshighat and Khadipakha were selected.

Banshighat, also called Gajendra Basti is located in KMC Ward 11, and Khadipakha is situated on the steep land of KMC Ward 3. These are one of the oldest informal settlements of Kathmandu Valley. There are altogether 136 households in Banshighat, and 366 households in Khadipakha mostly from Sindhupalchok, Nuwakot, Ramechhap, Sunsari, and Surkhet. Most of the residents are daily wage earners, street vendors, house maids, office helpers, etc. Some women are involved in weaving, tailoring and running a small grocery shop in their own house. According FGDs, their earnings are meagre, but still are able to save for their children's education.

3.6.1 Housing and Livelihood

Building Damage

The earthquake did not cause any significant damage to these informal settlements. Some houses developed minor wall cracks which were immediately repaired by the residents themselves. These houses are small, one or two storey structures with a CGI roof. These are semi-permanent structures and very few are properly constructed. Many are built with wood and plastic sheets and are one storey tall. They used open spaces nearby their settlements for temporary shelters.

Livelihood

Although there was no significant damage to the houses within informal settlements, there was a significant impact on their daily life due to loss of their jobs. The informal settlers were providing services to neighbourhood city dwellers as house maids, working in the shops, small factories, and street vendors. After earthquake, many city dwellers were displaced and closed their business due to damage to their houses. Due to loss of jobs and with no family income, the children could not continue to go to school after earthquake. During the FGD sessions, we discovered that informal settlers have had a difficult time for almost 6 months after the earthquake. However, the situation is improving. However, there were some cases of depression and high blood pressure after earthquake mainly due to fear of aftershocks.

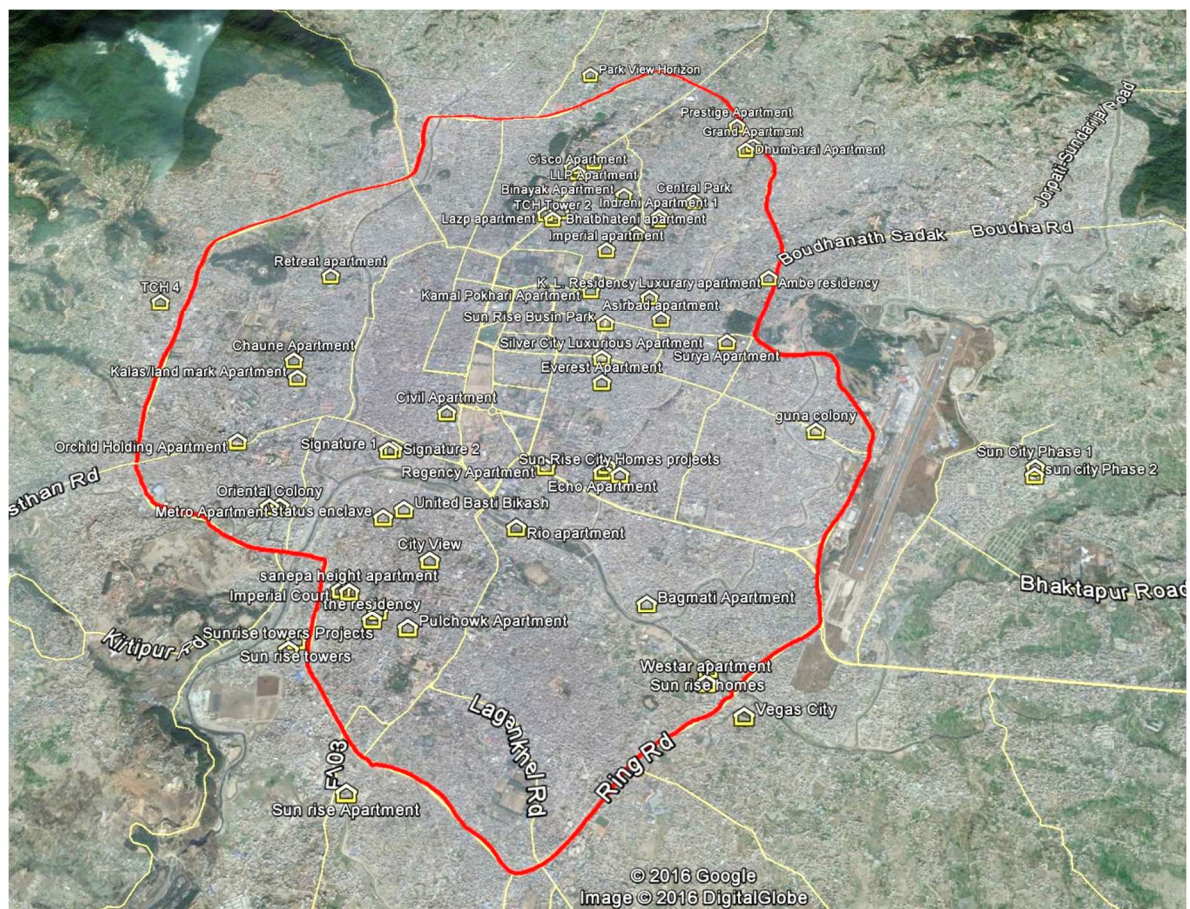
3.6.2 Cultural & heritage

In Banshighat, there are many monuments of heritage value which were severely damaged. However, the informal settlers of this area, who have migrated from other districts in Nepal, have no special ceremony related to these temples and monuments unlike those in core settlements of the valley. They celebrate all festivals like Dashain, Lhosar, Teej happily and joyfully which help them put aside their troubles, problems and pains of earthquake as mentioned during the FGD.

3.7 Apartment Buildings

High-rise apartments is a recent phenomenon in Nepal. In the past decade, high-rise apartments have emerged as an alternative long-term solution for urban inhabitants. Various study reports show that a substantial number of people are migrating from rural to urban areas for a better life and opportunities every year. One of the biggest challenges is how to accommodate these migrants in urban areas by optimally utilizing the available land. In this context, high-rise apartments have come up as one solution. However, the price of a unit is not cheaper than an individual house.

Fig. 3.3 Apartment Buildings in Kathmandu Valley



The Joint Housing Ownership Act 2054 (1997) and the Apartment Regulations, 2060 provided the legal and procedural basis for this new housing type. Since then a total of 71 permits for high-rise apartment construction have been issued to developers by the concerned government department. According to DUDBC, among these permits, 43 apartment buildings were completed and inhabited before the April 2015 earthquake, 7 were in the construction stage, 13 have changed their use, and the status of the remaining 8 approved apartment permits is unknown till the study period.

A total of 6,950 individual apartment units can be built given the 71 permits that have been approved. The apartments have on an average 98 units per apartment and range from 12 units to 600 units per building. Most of these apartments have an adequate water supply system.

Similarly, other basic facilities like sewerage system, electrification, road etc. are also provided. These units are generating comparatively higher revenues for government. Therefore, their contribution and importance for future urbanization perspectives cannot be overlooked. In order to assess the impact of earthquake on apartment buildings, Sunrise Apartment of Nakhu, Lalitpur was selected for FGD. This apartment was completed in 2071 BS with 262 units, of which 155 units were occupied at the time of earthquake. They were mostly financed by Bank with 20-30% self-finance.

3.7.1 Housing and Livelihood

Building Damage

None of apartments reported a complete collapse though many high-rise buildings suffered damage. About 4,348 families were living in 43 apartment buildings at the time of the earthquake. All of them were directly affected and displaced. While no casualties were reported, one person was found dead outside a building.

Rapid Visual Evaluation was conducted on 120 building blocks of 52 apartments including 9 that were changed to commercial buildings by the evaluation team of DUDBC. Two apartment buildings were marked Red - declared unsafe even for entry and residing. An additional 10 apartments were marked Yellow and restricted from immediate use. Another 32 apartments were marked Yellow and certain area were restricted. The remaining eight apartments were given green stickers with no restriction on use.

In case of Sunrise Apartment, there were many damages, i.e., cracks on walls, plaster, carnation carnes, wall and hair cracks in beams and pillars. The developer claimed approximately loss of three hundred million rupees equivalent to 3 million USD. On the other hand, the residents lost their valuable property inside the unit. At that time about 155 families resided in the apartment complex. Certain units which were funded by a bank loan were covered by insurance and had to pay an annual NRs. 4 per sq ft as insurance premium. However, the insurance policy covers only structural damages and common infrastructure, while interiors and physical assets are excluded. The developer agreed to repair all damages and take necessary responsibilities so that there is no additional fiscal burden on unit owners. The damage assessment has been done by the developer, a third party and the government. Damage repair is being seriously monitoring by the concerned government department, which will issue a clearance certificate after the successful completion of all the repair tasks. This would indicate that the apartment is safe for residing. However, at Sunrise Apartments, 110 units initiated their reconstruction of individual units and around 55 units already shifted in their units.

At other apartment complexes, after the rapid assessment, owners are gradually returning to their units after necessary repair work has been completed and green stickers have been issued. A few yellow sticker buildings have initiated repair work. Repair work is also possible as most apartments are insured. Repairs are dependent on the type of insurance coverage, joint or individual coverage, and responsibility of the developers.

During the FGD sessions, some mentioned that it might be wise to limit the height of these buildings to only a couple of storeys. There is a perception that small houses are safer for inhabitants and the surroundings. According to them, they would not go back to their apartments, if they had another option. They were open to selling it as well. Those, who had an alternate house or were capable of renting a new house, have already left the apartment. There

is a general fear regarding the fragility of high-rise apartments and their safety. During the FGD, residents were expressed the social pressure the felt from relatives and friends who were concerned for their safety after almost every aftershock.

The Gorkha earthquake has brought forth the realization of the need of separate open spaces for apartment dwellers as the ground allocated within the apartment is of no use in such a situation. The hundreds of people were not able to find safe open space nearby their apartment and had to live with their relatives at other locations.

4 Rescue, Relief, Recovery and Reconstruction

Rescue, relief and recovery are three important tasks during the post disaster management period. This part of the assessment analyses the rescue, relief activities, and the recovery initiatives taken by different stakeholders. It also assess the current reconstruction status in urban settlements till the study period. Most of the information provided in this part of the study are based on the information provided by FGD and KII.

4.1 Rescue Initiative

Rescuing victims from badly damaged buildings, specifically collapsed RCC frame structure buildings were very challenging in dense urban settlements. Government's security forces such as Nepal Army, Nepal Police Force; Nepal Red Cross; IOM, and local youths and rescue forces from different countries played important role in rescuing the earthquake victims from collapsed buildings and rubbles. International search and rescue workers worked in their neighborhood for rescuing the victim according to FGD and KII were Belgium Search & Rescue Team, Canadian Rescue Team, China International Search and Rescue Team, Israel Search and Rescue Team, Malaysia Search and Rescue Team and South Korean rescue workers. Participants in FGD highly appreciated Nepal Army, Nepal Police and the local youth volunteers' dedication in rescue activities.

4.2 Relief Initiative

Government and various agencies were involved in the distribution of relief materials in affected areas. The Government of Nepal provided cash, whereas other organizations mainly provided different relief materials to alleviate the victims' suffering. Relief supports provided were mainly tarpaulins, health kits, kitchen utensil, mattress, food stuffs, water purifying kits, temporary toilets and other sanitation items. Few of supporting agencies also provided CGI bundles and construction support for temporary shelters, in historic and market towns, and settlements of Thimi, and Bhaktapur in Kathmandu Valley core. The temporary shelters are still being used in these by earthquake victims in such settlements. Based on information of FGDs of all sectors, the number of support agencies were noticed more in historic and market towns compared to core and peri-urban settlements of the valley.

Organizations, Civil Societies and Forums, Local communities and clubs, etc. involved in relief management and distributions mentioned during FGD and KII are described below:

4.2.1 Government's Initial and Winterization Support

The government announced an initial support amount of NRs. 15,000 and further winterization support of NRs. 10,000 to earthquake victims, whose houses were completely damaged. The victims list was recorded in their municipality's affected victims list. Based on the information gathered through FGDs and KIIs, majority of victims have received both supports in all settlements. A few households (about 10%) in KMC Wards 19, 20, 23, 25 and LSMC Wards 11, 18, have not received either type of fiscal supports.

Majority of the participants in FGDs expressed that the initial support amount was extremely meagre and almost insufficient to avail/construct a basic temporary shelter. Majority of the victims in the Kathmandu Valley core and peri-urban areas used these amounts for daily

household expenses. Most recipients in historic towns and market towns used these funds in building temporary shelters, while a minority used it for incidental household consumption.

During the FGDs it came forth that the winterization support was mainly used for household consumption, while a few used it to pay back loans taken for managing daily expenses during the post-earthquake period. During winter, many victims managed with their existing set of winter wear, or got them from other support organizations. Renters and informal settlers could not benefit from such fiscal support schemes, due to the government's criteria that only the legal owner of the damaged house was eligible for such support. This myopic decision caused great misery to poor renters residing in damaged houses to be neglected by the government during such a national tragedy. The people living in informal settlement whose livelihood depends mainly on daily wages also suffered badly although they did not face shelter problem due to such criteria.

4.2.2 Ward Citizen Forum

Ward Citizen Forum (Woda Nagarik Munch) is municipality's ward level committee formed in coordination with municipality's ward office. A Ward Citizen Forum comprises of twenty seven members, representing political parties, civil societies, youth, communities, senior citizens, local youth clubs, senior citizen, single women, differently able people, NGOs, etc. Some of the Ward Citizen Forum have covered multiple wards. In most of the earthquake affected areas, these forums were actively involved in distributing relief materials received from various sources. These forums according to FGD and KII were very active, in exception to the settlements in Kathmandu Metropolitan City and Gorkha Municipality. Ward Citizen Forums are the key medium of reaching victims for delivering relief materials for DDRC, CDRC, corresponding municipalities and other stakeholders in municipalities. Members of ward citizen forum were the means of approaching community for FGD also for this study; they helped study team to arrange meeting place to conduct FGD and connecting municipalities and key informants for collecting necessary data and information for the study in case of Panga, Bungmati, Khokana, Sankhu, Bhaktapur. The study team experienced difficulties in connecting earthquake victims and key informants to collect information where these Forums are not very active.

4.2.3 DDRC and CDRC

DDRC and CDRC mobilized under Home Ministry's Disaster Management Section is one of the largest authorized committee for distributing disaster relief materials in disaster areas. The relief materials specially provided by government channel, Red Cross and other government and non-government agencies were delivered to the victims through these committees. These committees seem active in most of the study areas according to FGD and KII. They do not seem efficient in Kathmandu Metropolitan City according to FGD in the study area of Kathmandu Metropolitan City. DDRC / CDRC in coordination with Ward Citizen Forum distributed relief materials according to FGD and KII in most of the study areas.

4.2.4 Community Based Organizations & Local Youths

Community based organizations, local clubs and local youths played impressive role in relief and rescue in this earthquake disaster. They were actively involved in collecting and distributing relief materials, managing temporary shelter, rescuing earthquake victims from

rubbles, moving valuable things from damaged buildings to safe areas in neighborhood, safeguarding valuable things and important artifacts in damaged buildings according to FGD and KII. Lyama Pucha of Madhyapur Thimi, Jyapu Samaj of Lalitpur, Machhendra Club in Bungmati and Friends of Sankhu are some of the good examples of the active community based organizations. Youths actively involved in moving valuable artifact from the damaged private house and help in storing them in Itumbahal are good examples of youths' active involvement. Self-motivated initiatives of youths and local people in Kathmandu Valley not only limited to Valley but contributed beyond it to support earthquake victim this Gorkha Earthquake 2015. The participants of FGD were very grateful to local CBOs, youth and civil societies for immediate support during earthquake. Local youths and people have key role in relief management in case of the settlements in Kathmandu Municipality area and peri-urban areas where Ward Citizen Forum were not active.

Few other local community based organizations found working in historical towns, Kathmandu Valley core and informal settlement were Khokana Women Cooperatives for immediate relief support in Khokana, Shreejanshil Sramik Mahila Samuha for supporting rental and host families, Panga Women Coopeative in Panga for recovery, Liwali Ganesh Bukampa Pedit Punasthapana Kendra in Bhaktapur for recovery, Mahila Ekta Samaj in informal settlement for recovery, Nawa Astha Women Cooperative for recovery in Chyasal.

4.2.5 Other Organizations

Many National and International Agencies were also involved in Relief initiatives. Some of the key organizations and their support identified on the basis of FGD and KII are provided in the Table 3.1. Religious groups such as Budhhists, Christian, Muslim and Hindu also supported in immediate support to the earthquake victims. The number of support organizations are noticed few in the Kathmandu of core and peri-urban areas compared to other areas based on the information provided by FGD and KII.

Table 4.1 Organizations Involved in Rescue & Relief Support

Sector	Supporting Organizations
Kathmandu Valley Core	Immediate relief - Chaudhari Group, Himalaya Foundation, Italian Institute, J-RAPID program, Madesi Janadhikar Forum, Marwadi Sewa Samiti, Nepal Red Cross, Rotary WASH – ENPHO, OXFAM Temporary Shelter - Chuche Foundation, CIUD, Lumanti, UNHABITAT
Market Towns	Immediate Relief - CDF Nepal, CFF Nepal, Nepal Red Cross, Samaritan Purse, SEBAC Nepal, World Food Program, World Vision Temporary Shelter - UNHABITAT
Historic Towns	Immediate Relief - ACTED Nepal, Asia Child Support, CDMC, Change Nepal, Red Cross, FNNCI, Indian Embassy, Islamic Relief, Jai Nepal, Kathmandu University, Korean relief team, Korean Samaj, Lions Club, Marwadi Sewa Samiti, Rara Hill Foundation, Rotary Club, Sikh Aid, Turkis relief team, World Vision WASH - ENPHO, OXFAM; Temporary Shelter - Action Aid, CIUD, Danish Project, Lumanti, UNHABITAT
Peri-urban	Nepal Red Cross

4.3 Recovery Initiatives

Recovery is one of the most challenging tasks in post disaster management in the devastating disaster like Gorkha earthquake. If the decisions and actions for the recovery are not taken properly, the damage caused by this could be worse than the damage made by actual disaster. On the other hand, the disaster can turn into opportunities if the recovery plan can analyse the situation of the victims in pre-disaster and manage the recovery plan properly in building back better (BBB) communities. National Reconstruction Authority along with government organizations, NGOs, INGOs, local CBOs, Civil Societies and affected communities themselves are the responsible actors in transforming disasters to opportunities for building back better communities. This part of the study analyses the undertaken activities and initiatives by these actors in selected urban settlements based on FGDs, KIIs and secondary data collected from various sources.

4.3.1 Government's Financial Scheme for Recovery

Government of Nepal has announced refinance facility for concessional home loan scheme to support earthquake victims through Banks and Financial Institutions (BFIs) for 5 years in minimum to maximum of 10 years period. Government will provide refinancing a maximum loan amount of NRs 2.5 million to affected families inside Kathmandu valley and outside the valley the limit is NRs 1.5 million. The eligible earthquake victims will get this loan from BFIs at 2% interest rate. BFIs are not allowed to levy any other charges except insurance premium, valuation fee, credit information fee and loan guarantee fees. The BFIs will disburse loans to only those who have a satisfactory income to pay back the loan and can provide adequate collateral. Loan will be disbursed in at least four installments based on the progress of the construction. The engineers deputed by the Government through NRA in corresponding municipalities of the settlements will supervise the construction progress and approve for the installments. The four installments are - (i) after acceptance of loan disbursement, (ii) after receiving building permit up to DPC, (iii) after completing DPC and getting permit for super structure, and (iv) after receiving building completion certificate from corresponding municipality.

The government has also provided alternative provision of NRs 200,000 financial grant with up to NRs. 300,000 microfinance concessional home loan under this refinance scheme. This loan will be disbursed either through microfinance bank or other BFIs for eligible earthquake victims, and considered as deprived sector lending. One who apply for bank loan as of the government scheme cannot apply for the grant.

The study team has analysed if the victims were aware of government's initiation for financial support and their perception on the government's financial scheme and the current status of using the fund. The FGDs analysis shows that the earthquake victims were aware of these schemes but were not clear about the mechanism of getting any of these supports. None of the participants know anyone who have succeeded to get any of these funds. The study also tried to identify if any BFI visited to the settlements to make them understand or counselling victims in getting the fund from BFI for reconstruction. But none of BFIs has visited with scheme or plans in any of the studied settlements till the time of survey.

During FGDs in valley core, historic towns and peri-urban settlements, the participants were expressing dissatisfaction of Rs. 2 lakhs grant as this amount is not sufficient even to remove

the debris or dismantling the damaged houses in city areas where the cost of labour and transportation is very high. Further, they have assumed that there would be more expenses on fulfilling all requirements in the complex process of getting the amount at different installments.

Regarding the Rs. 2.5 million loan, the participants were willing to get this support, but majority of them were not optimistic due to limited source of income which is main eligible criteria of this scheme. Most of earthquake victims of the selected settlements lie in the low income group especially in the Kathmandu Valley core and historical towns. During FGD, an earthquake victim living in temporary shelter shared his experience of applying for loan Rs 2.5 million through a commercial bank, but was failed to proceed further as his monthly income of Rs 15,000 was not sufficient to be eligible for the loan.

4.3.2 Community's Perception in Traditional Buildings

Through FGD the study team has tried to get the perception of local communities towards their interest to build traditional buildings. Although the participants were aware of the need of maintaining the traditional architect of buildings, they were not ready to build such buildings on their own because of high cost of construction compared to RCC frame buildings. Such perceptions were mostly noted in valley core and historic towns where large number of houses with traditional architect were damaged. They are ready to maintain traditional architecture if government provides some support for the reconstruction of such houses. A participant in FGD in Sankhu had a traditional house with artistic window, which was seriously damaged by the earthquake, and now she got confused of this artistic window whether to sell it or repair as it is costlier to repair. During a field observation, a traditional Newari house with eight carved wooden pillars on the ground floor was found to be under reconstruction. The house owner have already spent NRs 1.6 lakhs only to repair these pillars. According to him, each wooden carved pillar costs about NRs 20,000 and he wanted to reconstruct his house to its pre-earthquake state.

4.3.3 Community's Interest in Planned Settlement

A planned settlement is essential for sustainable development. Recovery of an individual buildings alone may not support to sustainable development of the settlement. Recovery the damaged houses and other infrastructure in the earthquake affected areas is the great opportunity for recovering the settlements once destroyed by unplanned urban development. The recovery of the settlements should recovery of economy, social and cultural, natural, tangible and intangible heritage, building and other infrastructures. In this regard, the study has analysed if the communities are interested in developing planned settlement especially in valley core and historic towns and also to identify the community's initiations in developing planned settlements.

Many participants of FGDs in Kathmandu Valley core and historic towns have shown their interest in the planned development but they do not have idea how to proceed and where to contact. The study has identified few communities which have initiated in settlement level planning – (i) plan of restoring private houses in Pilachhen, Lalitpur, (ii) Plan to go for house pooling in Dambo Chowka, Kilagal, (iii) Plan to restore private house in Jela Bhaktapur, (iv) Master Plan for development of Khokana, prepared by local technical people and (v) Bungmati Reconstruction Plan with technical support of UNHABITAT (Annex 4.1). Pilachhen has

already started constructing two houses. The study team visited the development site and discussed with Mr. Ramesh Maharjan, the key initiator to understand their concept and the current status. The team also discussed with the president, Mr. Macha Raja Maharjan, Dambo Chowka in Kathmandu. Besides above lists, communities in Mussun Bahal, Kamalachhi, Suchika Galli are in initial phase for house pooling and restoring private house, while Piganani is exploring the possibilities of traditional style reconstruction.

4.3.4 Government's Initiation in Safe Building Construction

MoFALD along with the Ministry of Urban Development came up with a "Settlement Development Guideline (SDG)" after earthquake to control haphazard construction of buildings and instructed municipalities for its strict implementation. As mentioned in KIIs with authorities of municipalities in selected urban settlements, they have already implemented NBC and SDG.

Government and other organizations such as UNDP, USAID, NSET are providing mason trainings in order to develop quality human resources so that the construction quality can be maintained. Government has recently recruited more than 1200 engineers to support in designing and supervising building construction work in all 14 affected districts. This study has also analysed the current status of practice of taking building permit, implementation of the building code, using trained masons, etc., which are discussed in Chapter 5.

4.3.5 Role and Capacity of Municipalities

The Local Self-Governance Act 1999 has provided an extensive and wider power to municipalities by allocating huge range of activities and duties in their municipal areas. They have to carry out various activities related to finance, urban development, water resources, environment and sanitation, education and sport development, culture, works and transport, health service, social welfare, industry and tourism, and other miscellaneous works. They can prepare, implement and monitor their own plans.

The capacity of municipalities in the earthquake damaged areas vary in terms of their technical capability, financial and managerial capabilities. However, they have been facing similar types of problems in reconstruction and recovery process in the post-earthquake period. First, they are yet to come up with clear vision and reconstruction plan for the earthquake damaged area integrating individual houses into integrated infrastructure development. Despite clear guidance from the MoFALD and MoUD's directives for preparation of necessary legislation and guidelines for house pooling, none of them (whether big or small municipalities) has prepared such regulations so far. Second, all of them are confused in issuing building permit for the extension or addition of floors in the existing houses, as the provisions of column size and other requirements were different in the pre-earthquake situation. One needs to construct at least 300 mm X 300 mm minimum size of RCC structure, irrespective of room space now whereas such provisions were not there before the earthquake. The municipal staffs get confuse how to issue permit for those household who came for floor extension with RCC column less than the prescribed size.

In addition to these, many facts demonstrate their inadequate capacity. First, rapid urbanisation causing growth of settlements, construction of infrastructure and individual buildings particularly in the Kathmandu valley from the past two decades have hardly considered the existing geological map, prepared by Department of Mines and Geology with support from Federal Institute for Geosciences and Natural Resources, Germany in 1998. Not only individual

houses, but some high rise apartments have also been already built in those exposed and potential liquefaction areas. Even the steeped sloped, along the riverbanks and land fill sites have been used for the construction of individual houses and row housing. There has been a 'Kathmandu Risk-Sensitive Land Use Planning' prepared as a part of Earthquakes and Megacities Initiatives with support from German government in 2009. Recently, the KVDA with support from JICA has prepared another risk sensitive land use plan but its implementation is not yet done. Second, with increasing the number of building permit application due to rapid urbanisation, there are no proportionate growth of human resources in the municipalities. Recently, most of the municipalities in the earthquake hit areas have reopened the building permit application after holding for many months after the Gorkha earthquake. According to municipal sources, there has been significantly increase in building permit application after Dashain and Tihar, the best seasons for construction. Also, many households do come to municipality for getting the 'completion certificate' of their houses. For the site verification, municipalities have hired 'verifier' who are in most cases non-technical persons. In Kathmandu Metropolitan City alone, it receives 10-15 application for building permits each day. However, Tokha municipality receives only 2-3 application for building permit each day. The corresponding figure for Karyabinayak municipality is 5-6 application each day.

In Kathmandu Metropolitan City, there are only few number of 'verifier.' As a result, issuing building permit is unnecessarily delayed. For the case of Bhaktapur city, one person has to cover two wards to check the drawings in the construction site. Third, no municipality of the valley has yet managed to implement the full cycle of building regulation by establishing (a) prior approval, (b) inspection and (c) enforcement/penalties. Enforcement of the National Building Codes to ensure earthquake resilience for new private construction remains a major challenge for all municipalities, while retrofitting of existing private buildings has simply not begun.

Municipalities visited during the project period identified that there is no any specific reconstruction plan in municipalities under this study, hence no any budget has been allocated for reconstruction in these municipalities.

4.4 Status of Reconstruction

The status of the reconstruction determines the status of the initiations taken in the process of recovery from the disaster. The status of reconstruction help concerned agencies to correct their plans and programs or to change their strategies to achieve their goal timely. All municipalities and VDCs in earthquake affected districts stopped issuing building permits for some months after earthquake to avoid the risk of occurring more aftershocks and their possible impact in new buildings. Government revised the permit standard and came up with new Settlement Development Guidelines to build resilient buildings for municipalities to follow. However, the reconstruction process is found very slow as noted by technical survey and the number of building permit issues by municipalities in the surveyed urban settlements (Table 4.1). It is important to note that in some settlements like Thimi and Panga, there were more buildings under construction than the number of permit issued by corresponding municipalities. Similalry, in Bungamait, Khokana and Chautara, there were few buildings noticed under construction although building permit was not issued. This implies that people are still practicing construction without building permit, so there is high risk of illegal construction, if not monitored properly.

Table 4.2 Status of fully damaged houses and construction

Sector	Settlement	Fully damaged houses			Permit Issue ³	Construction ⁴
		Total houses ¹	Identified in field ²	%		
KV Core	Kathmandu	5104	1399	27%	94	76
	Lalitpur	3278	775	24%	131	71
	Bhaktapur	5950	1421	24%	NA	27
	Thimi	788	245	31%	21	26
Peri-urban	Gongabu	150	8	5%	NA	10
Densely settled	Ghattekulo	25				
Historical Town	Kokhana	812	119	15%	0	1
	Panga	503	135	27%	22	30
	Sankhu	1420	208	15%	NA	7
	Bungamati	989	92	12%	0	1
Market Town	Chautara	899	96	7%	0	7
	Charikot	808	16	11%	NA	1
	Gorkha	653	21	3%	47	10
Total		21379	4535	21%	315	277
¹ Fully damaged houses was estimated based on all wards included (complete or part of the ward) in the selected urban settlements and included both already demolished and not demolished damaged houses, Source: Concerned municipalities. ² Houses identified were undemolished fully damaged houses noted during Technical Survey ³ Number of building permit issued by corresponding municipality in the wards included (completely or partly) in the selected settlement, Source: concerned municipalities, permit issued till April, 2016 ⁴ Buildings found under construction started after earthquake during field survey, till April 2016						

The study has analyzed the reasons behind not starting the reconstruction processes through FGDs. The participants were asked for their opinion on six major reasons - (i) Lack of financial resources, (ii) Lack of technical skills or guidance, (iii) Legal issue of land (iv) Building permit not issued, (v) Waiting for government policy/decision, and (vi) Risk of dismantling neighbours' building; reasons were prioritized in order from 1 to 6, with 1 the most important.

Table 4.3 Reasons for not reconstructing

Reasons for not constructing	KMC	LSMC	THIMI	Bhaktapur	Valley Core	Panga	Khokana	Bungmati	Sankhu	Historical Towns	Chautara	Charikot	Gorkha	Market Towns	Peri-urban	Overall
Lack of financial resources	2	1	1	1	1	1	1	1	1	1	1	2	1	1	1	1
Waiting for government policy/decision	1	2	2	3	2	5	3	6	3	5	2	1	2	2	2	2
Legal issue of land	4	6	3	4	4	4	4	4	2	3	3	5	3	3	4	3
Risk of dismantling neighbours' buildings	3	3	5	2	3	3	2	2	5	2	6	6	6	6	6	4
Lack of technical skills or guidance	6	4	4	5	5	6	5	5	6	6	5	3	4	3	3	5
Building permit not issued	5	5	6	6	6	2	6	3	4	4	4	4	5	5	5	6

The “Lack of financial resources” is the first reason of not reconstructing followed by “waiting for government policy/decision”, “Legal issue for the land”, “Risk of dismantling neighbours’ buildings”, “Lack of technical skills or guidance”, and “Building Permit not issued” in overall scenario. The reason “Lack of the financial resources” stands as the first most important reason for all sectors, but other reasons are different in priority order for different sectors. Valley core, market town and peri-urban sectors have considered government policy as the second reason whereas historical towns ranked it to the 5th. Risk of falling down the neighbour’s house while dismantling their own house is ranked second by historic towns, which is ranked in third by KV core, but others ranked it in 6th. Similarly different settlements ranked differently as of their priorities (Table 4.2). The existing condition of debris, undemolished houses, and building under construction are briefly explained below.

4.4.1 Debris Clearance

Debris clearance is one of the initial tasks of reconstruction, this is also a very challenging in case of densely settled and narrow lane areas specifically in KV core and historical towns where many buildings are damaged and have many narrow streets. Debris clearance work is still remaining in most of the settlements in KV core and historical towns obstructing the vehicle movement, especially in Khokana, Panga, Kathmandu, Bhaktapur and Thimi. From the FGD. Municipality in cooperation with other line agencies cleared the debris in main streets, squares, and important public areas but have not supported in clearing the debris of private owned buildings. In some settlements like Sankhu, IOM was clearing the debris, but facing problems

of permission in case of buildings whose owners have already migrated before or after Earthquake. Due to lack of land ownership documents and conflict between neighbouring attached houses, the owners were reluctant to clear debris of the foundation in fear of missing the boundary of their houses, especially in Khokan, Sankhu and Panga. IOM has significant involvement in clearing debris in market towns according to FGD & KI, which was also observed in few towns during field work.

4.4.2 Status of Undemolished Buildings

The study team has tried to gather some qualitative information on the status of reconstruction in selected urban settlements during FGDs. The basic six statuses were characterize existing condition of damaged houses in the settlements and noted their responses as – none, few, some, many, and all- buildings in each FGD, (Table 4.3). Upon analysis of all FGDs’ responses, the average response was presented in the table. It

Table 4.3 Size of damaged houses by status of reconstruction				
Status of reconstruction	Average response of communities on size of damaged houses			
	KV core	Peri-urban	Historic Town	Market Town
Completely demolished but not yet started	Some	many	some	some
Only top floor demolished and foundation remained	Some	many	many	some
Reconstruction started or completed on the same foundation	Few	some	few	none
Completely demolished and reconstructed with new foundation	Few	none	few	none
Damaged building with temporary support (<i>teko</i>)	many	some	some	none
Damaged houses are still being used	many	many	some	none

shows that in Kathmandu Valley core, many damaged houses are supported by teko and are still being used by victims, some damaged houses were demolished only top floor and some completely, and few houses were reconstructed on the same foundation and some with new foundation. This scenario is not same in all sectors as shown in the table. However, this analysis was consistent with the data collected from Technical survey of undemolished and newly constructing buildings.

A quick field survey conducted in the study area has identified 4,535 undemolished houses. These buildings were marked as not appropriate for living during damage assessment, but are still standing in the settlements without demolishing or restoration resulting high hazards to densely populated neighborhoods, mainly in Kathmandu Valley core and historic towns. This is 21% of the total 21,147 completely damaged houses. The total number of 21,147 damaged houses are the aggregate of the damaged houses according to the data provided by corresponding municipalities. It is important to note that the damaged houses are aggregated in ward level, but the recorded houses are only belongs to the ward that includes in the selected settlement, for example ward 12 and 29 in case of KMC (Annex 1). The estimation may be little less in ward which is not included completely within the study area. Rate of undemolished houses in the core settlements are higher than in other settlements, and lowest in peri-urban area.

4.4.3 Safety Measures of Undemolished houses

The survey shows that only one percent damaged houses were completely restored, another one percent houses were partially restored, and about 98% damaged houses are in need of safety measure. There were about 37% damaged houses without any safety measure, 32% houses have reduced some level of risk by dismantling partially and 30% damaged houses were supported by just temporary support (*teko*), Fig 4.1. The details of status of undemolished houses by sector is shown in shown Fig 4.2 for four sectors and for all urban settlements in Fig. 4.3.

Fig. 4.1 Safety Measures of Undemolished houses

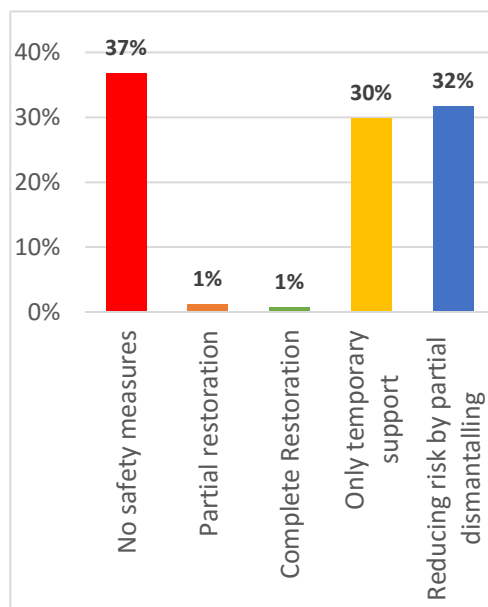


Fig. 4.2 Safety Measures of Undemolished Houses by Urban Sector

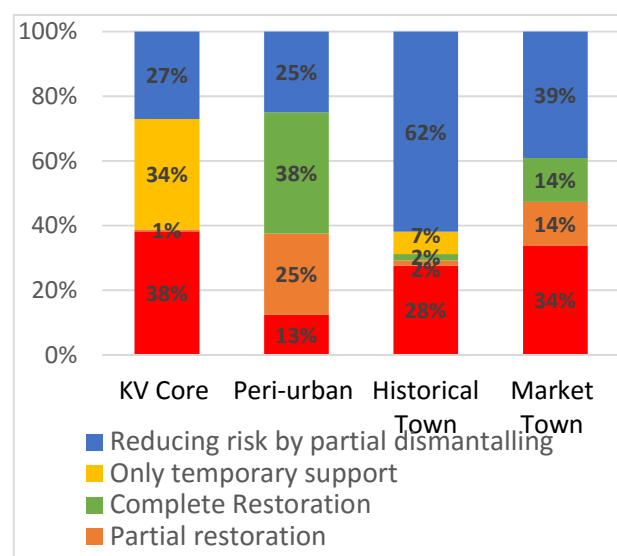
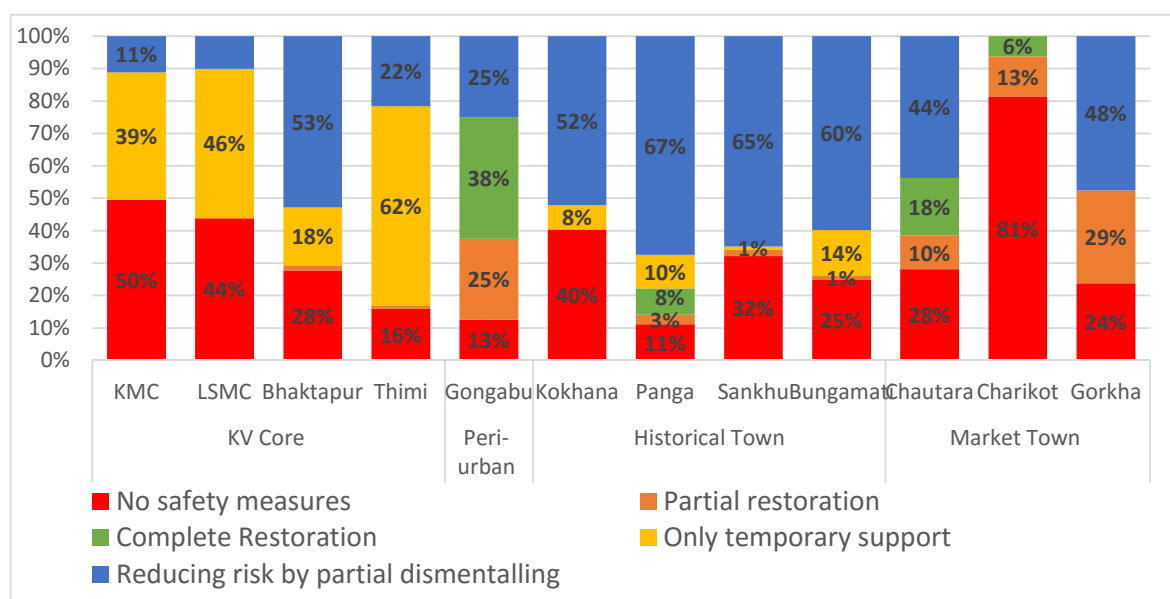


Fig. 4.3 Safety Measures of Undemolished Houses by Urban Settlement



4.4.4 Use of Damaged Houses

The graph in Fig. 4.4 and 4.5 show that the damaged and undemolished houses are still being used by house owners and renters, this may be due to the lack of alternatives and unaffordability for renting the house. The technical survey has noted that more than 50% of damaged houses which are not yet dismantled and were still using in the urban core settlements and historical towns (Fig 4.4). In case of peri-urban and market town, these houses were using after complete or partial restoration or dismantling the upper floor reducing the risk. In some settlements, these

Fig 4.4 Status of use of undemolished damaged houses

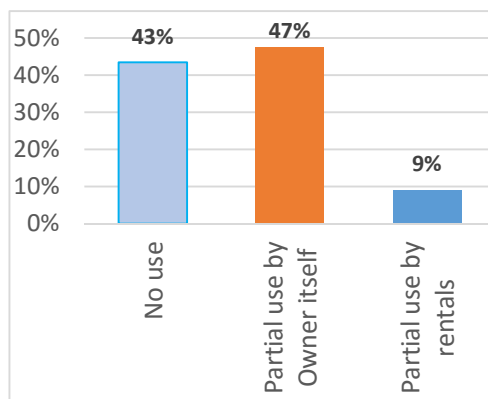
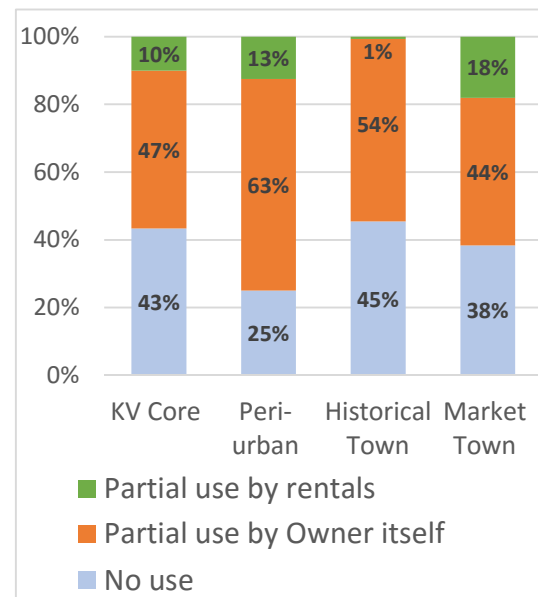
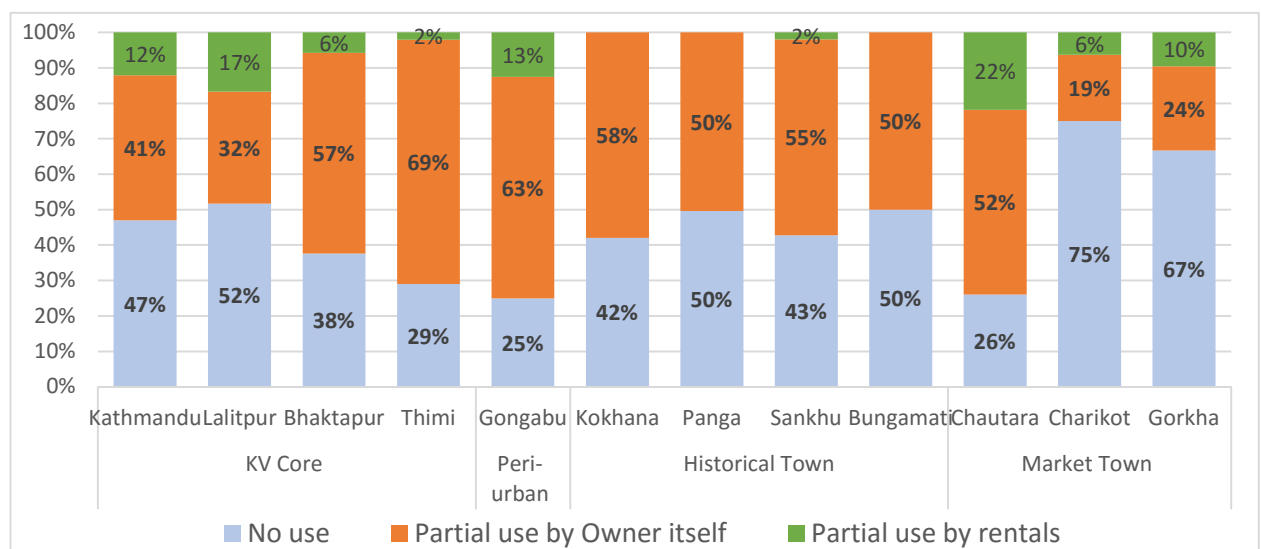


Fig 4.5 Status of use of undemolished damaged houses in urban sector



damaged houses were being used by renter only. In total, about 43% houses identified in the field were not in use (Fig 4.5). The details of undemolished damaged houses identified during technical survey in selected urban settlements are shown in Fig 4.6.

Fig 4.6 Status of use of undemolished damaged houses in urban settlements



4.4.5 Restoration of Damaged Houses

The technical survey has also attempted to identify if there were houses restored using different techniques. Few houses especially in peri-urban and market towns were found restored. Sixteen houses in total, in the market areas of Charikot and Chautara were found restored during survey, in which only one house was properly retrofitted, 11 were found jacketed with design, 11 were found jacketed, two were injected and two other technologies. The risk without design, 2 injected and 2 other technologies. The risk has not been reduced even after restoration because of restoration in inappropriate ways.

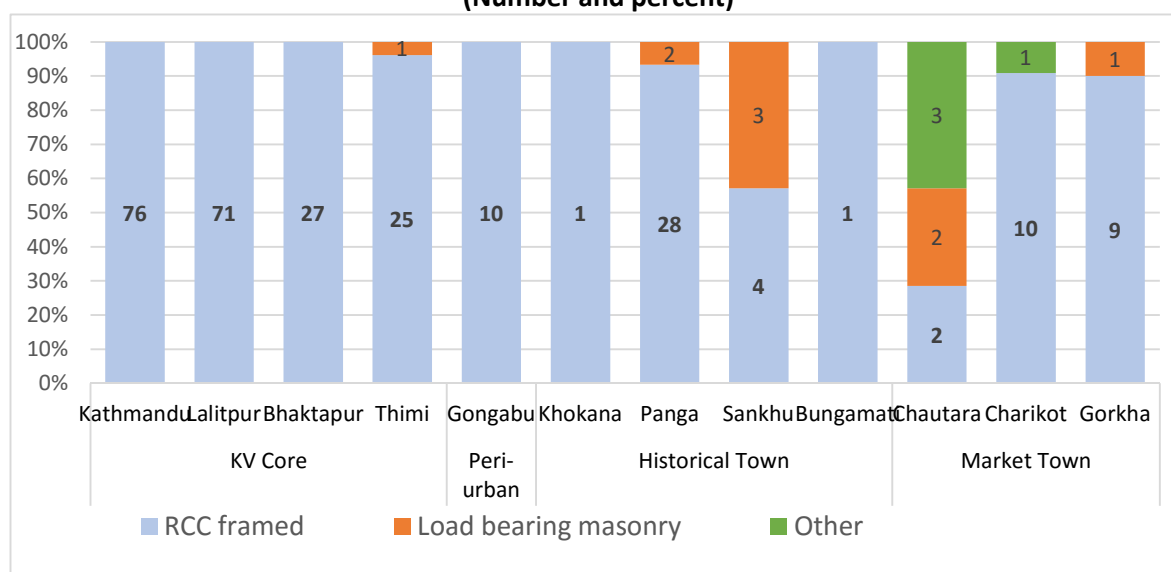
4.5 Status of Buildings under Construction

The technical survey has recorded 277 newly constructed buildings in 12 surveyed settlements, of which the highest number of newly constructed buildings were recorded in Kathmandu core, Lalitpur core, Panga, and so on (Table 4.1). The survey team has observed overall status of those buildings that include important features of engineering design such as type of structure (e.g., RCC frame, Load bearing), building level category, column size, use of earthquake resilience trained masons, purpose of use of building, etc. This is briefly explained below.

4.5.1 Structure Type of Reconstructing Buildings

Among 277 newly constructing houses, 95% houses were of RCC frame and only 5% were of load bearing and other. The percentage of newly constructed buildings by type of structure for the 12 settlements is shown in Fig 4.7. In exception to a building in Thimi, all buildings in KV core and Peri-urban settlements are RCC frame structure, this implies people in these areas preferred only RCC frame structure. RCC framed structure buildings have dominated other types of buildings also in historic market towns (Table 4.4), which validate the people's preference shown for going into RCC during FGD.

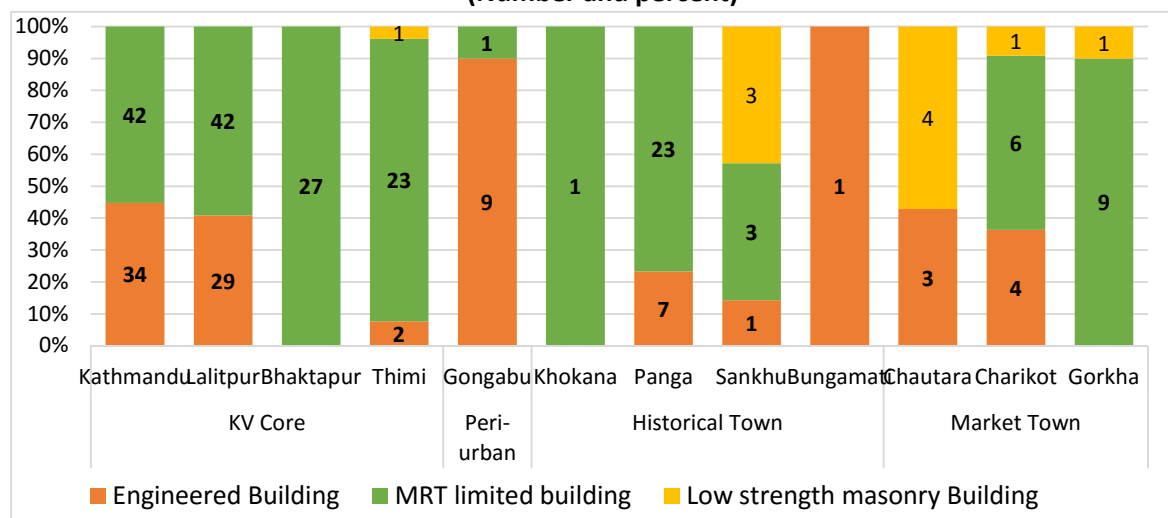
Fig 4.7 Type of structure of newly constructing building in urban settlements (Number and percent)



4.5.2 Building Level Category of Reconstructing Buildings

The technical survey has tried to identify the building level category such as state of art, engineered design, MRT, and low strength masonry. Of the total observed buildings, about 64% were MRT buildings, 32% were engineered buildings, 4% were found low strength masonry buildings, and none of them were state of art buildings found being constructed till the survey time. It is important to note that engineered buildings are dominant in peri-urban area, 9 out of 10 that is 90% are engineered buildings in these settlements. The details of buildings level category of newly constructing buildings for all 12 urban settlements are shown in Fig. 4.8.

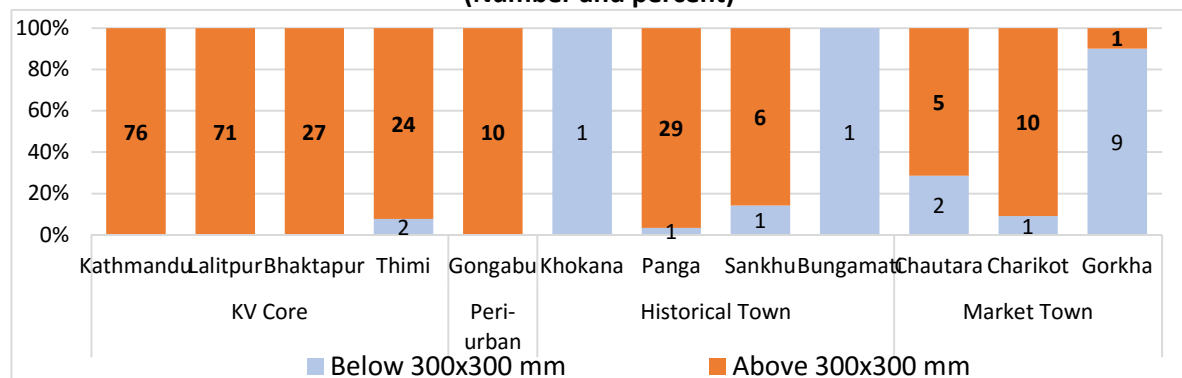
Fig 4.8 Building level category of newly constructing building in urban settlements (Number and percent)



4.5.3 Column Size of the Reconstructing Buildings

The study found 94% of total buildings under construction have followed the latest building code requirement of column size of above 300x300mm (Fig 4.9). The rest 6% have below 300x300 column size. People are still practicing RCC structure without fulfilling code requirements, especially in Market Town.

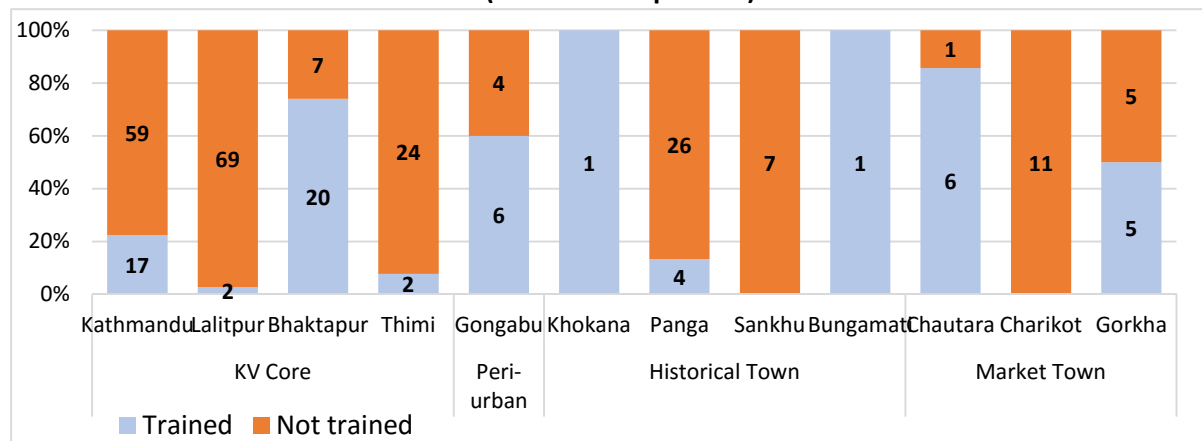
Fig 4.9 Column size of newly constructing building in urban settlements (Number and percent)



4.5.4 Use of Trained Masons in Reconstructing

The technical survey recorded that only 23% buildings were using earthquake resilience trained masons as shown in Fig 4.10, whereas government and many other organizations such as USAID/NSET, UNDP are providing masons training. According to the past studies and people in constructing business, there is high demand for trained masons but they are not available in the market. FGD in market towns reveal that most of the masons' trainings were focused to rural areas, local masons in urban settlements have less opportunity for earthquake resilience training.

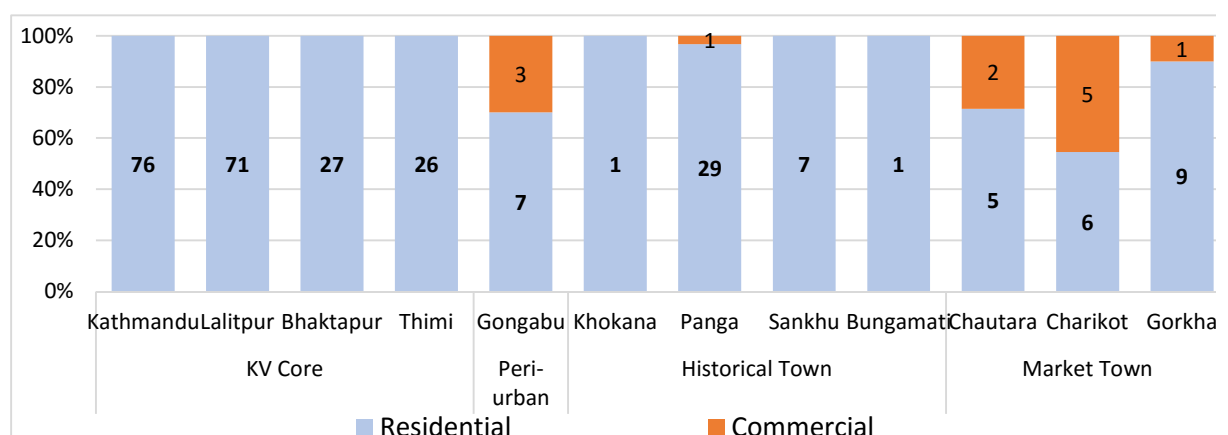
Fig 4.10 Type of masons used in newly constructing building in urban settlements (Number and percent)



4.5.5 Purpose of Using Reconstructing Building

According to building by laws, structural design of building is based on the purpose of use of the buildings. During survey, about 96% building were being constructed for residential purpose and only 4% buildings were for commercial purpose. Almost none of the buildings surveyed in Kathmandu Valley core and Historic Towns are built for commercial purpose, whereas 30% in case of peri-urban and 40% in case of market towns are built for commercial use as identified during field survey (Fig 4.11). The change of utilities should be controlled efficiently, chances of changing use in peri-urban and market towns is very high. Most of the buildings collapsed causing large number of death and casualties in peri-urban area and market towns were built for residential purpose but were using for commercial purposes.

Fig 4.11 Purpose of use of newly constructing building in urban settlements (Number and percent)



4.5.6 Building Permit Status of Buildings under Reconstruction

The study tried to analyse the status of the building which are under construction, only in Market Towns, the team can get such information. Such information collection was possible only in market towns of the 28 buildings under construction only 17% received building permits, 43% are in the process, 18% are preparing for application and 3% just ignored the permit. This shows the ineffectiveness of Building Permit System in the market towns, people started construction without approval. This could also be possible in other settlements.

4.5.7 Soil Vulnerability of the Settlement

People do not have knowledge on soil vulnerability of the settlements or they do not know whether the settlements are in safer place. Gorkha earthquake damaged several newly built RCC frame structure buildings in Peri-urban and Market Towns. Soil vulnerability could be the key issues in these settlements, this has to be incorporated clearly in building permit process.

4.5.8 Other Technical Requirements of Reconstruction

Some other technical requirements for the newly constructed buildings for safer building construction (as listed in Table 4.4) were also verified during field observation. The table provides the number of buildings identified in the field that do not meet the specified requirements in the left column of the table. For example, there were 178 buildings under construction had not followed 100mm stirrups spacing in confine zone in beams and columns. This indicates that after such a big earthquake, there are still buildings being constructed without meeting even basic technical requirements of safer buildings of monitoring too.

Table 4.4. Number of buildings not meeting other technical requirements in new constructions

Technical Requirements	No. of Building not meeting Technical Requirements				
	KV Core	Peri-urban	Historic Town	Market Town	Total
No soft storey effects	50	5	5	0	60
No short column effect	0	0	0	1	1
No eccentric connection of Irregular Shape	0	6	0	10	16
Grid beam connected to column	0	0	0	1	1
100 mm Stirrups spacing in confine zone in beams and columns	172	0	0	6	178
Main rebar-not less than 4 to16 + 4 to12 on ground floor	0	0	3	1	4
Main rebar-not less than 8 to12 on other floor	0	0	3	1	4
Horizontal tie band provided?	59	7	19	4	89
Tie band rebar connected to columns?	60	7	19	2	88
No room walls in projected cantilever	2	1	1	2	6

5 Major Issues and Long Term Impacts

This chapter explains about the issues and challenges identified based on various implication of Gorkha earthquake on housing and livelihood. These issues and challenges are need to be addressed in the recovery and reconstruction process.

5.1 Major Issues

The study team has analysed major issues faced by urban population. Some of the major issues are discussed below.

Debris management

Debris management has become an issue in the core settlements and historic towns in fringe. The local community with support of the concerned municipality were able to clear the main access after the earthquake. Recently, some routes are cleared off for festival routes. However, many dilapidated houses along the pedestrian alleys and inner courtyards are yet to be cleared off. House owners' poor affordability, lack of vehicular access to the houses and absence of financial support from the concerned municipality have hindered the debris management.

Dismantling of damaged houses in the core settlement and historic towns in fringe

Both traditional brick masonry houses and some RCC structure buildings damaged by the earthquake in the core settlements and historic town areas yet to be dismantled. These structures are creating high risk to the pedestrians and the occupants. Dismantling these damaged and vulnerable structures can easily pull down the adjacent attached structures, as many traditional houses have common walls with the adjacent buildings. Lack of equipment and technologies to dismantle RCC structure without damaging the surrounding houses and quick debris management without disturbing the narrow lanes and pedestrian paths are other two major issues.

In some cases the Government used bulldozer to remove the debris, such initiatives were detrimental particularly to the settlements where they have heritage valued possibilities of salvaged arts and artifacts.

Huge damage of unreinforced masonry construction in the core and historic towns and RCC failure in peri-urban and market towns

In fact, significant percentage of damages of old houses in the core settlements of the Kathmandu valley as well as in the historic towns in the fringe areas. Traditional houses of the core settlements and historic towns in the fringe are weak against the seismic risk due to shallow foundation, absence of damp proof course, poor bonding of mud mortar between the facing brickwork and inner brickwork, of different size and lack of tie at corner walls, and poor strength of building materials. Vertical division of houses and then haphazard renovation, alteration and addition of RCC floors on the existing old wall structure and then lack of consultation of professional and employment of trained masons in such activities are the major causes of such heavy destruction. Numerous reasons for collapse of RCC structures in peri-urban area of the valley include illegal construction, non-engineering construction with poor quality of workmanship during construction and change in building use.

Households living in vulnerable structures with inadequate safety measures

The survey of the damaged building status in the study areas confirmed that majority of damaged houses are yet to be strengthened through proper retrofitting. In the core settlements and historic towns in fringes, reduction of building floors by dismantling the upper RCC slabs and keeping temporary supports of wooden posts and iron pipes from outside as well as inside of the houses are two major measures carried out to avoid immediate risks. In both cases, about 60% of such structures are occupied either by home owners themselves or being rented out. In the case of KMC and LSMC, half of such houses are being occupied by house owners and renters. While restoring the damaged structures in 'Chautara,' jacketing is being used but without engineering design. As a result, risk is not reduced even after restoration of those structures.

Illegal Construction and Weak Monitoring System

Different forms of violation of building bylaws especially residential buildings have been seen particularly in Kathmandu and Lalitpur. There are general tendency of not getting building permit for addition and extension of the existing buildings. Others submit the blue print to municipality as per bylaws but violate the approved design and detailing. According to KMC, there are about 30% of the total cases of alternation and violation of the approved construction maps, most of them are of residential houses (Awale, 2014). Lack of separate and strong monitoring system within municipality and many clauses of the bylaws that do not fulfill present day needs of the people are the two major reasons for poor enforcement of the bylaws in the valley. Implementation of the National Building Code (NBC)³ has also been found to be ineffective. Although it has been made mandatory through the directives of the Ministry of Federal Affairs and Local Development, the Building Act 2055 BS and Building Byelaws 2064 BS have not included NBC provisions. As a result, it is yet to be included in the planning and building permit system. Even after 20 years of preparation, only few municipalities in Nepal are implementing the NBC along with building permit system. The implementation of architectural design requirements and planning guidelines (NBC 206:2003) was introduced in 2003 but the actual design has not been checked for compliance with this code. Many building materials available in the market and used in building construction are yet to be included in the building code and standards.

The weak implementation of building bylaws and the national building code in the valley has been well demonstrated in the recent earthquake. In KMC alone, there are many single buildings that collapsed killing more than one dozen people (Table 5.1). The reason of building collapse was ignorance of the building bylaws and failure to comply with the NBC.

The seven storey house at 'Sattale' in KMC 6 housed a motorbike workshop and Hair Saloon on the ground floor and a church in the sixth and the top floors. Seven families rented the middle four stories. This building collapsed in the first earthquake on 25 April 2015, killing 30 people, injuring 34 and the rescue of the remaining 26 persons by the Nepal Police. This house, covering a plinth area of 51.2 sq. m. in a 194.00 sq. m plot, was constructed in 1997. The building permit received was for a five storey house but the house owner built a seven storey structure. Built for commercial purposes with poor quality of construction, this house was sold some 6 years before (Himal, 2015). Implementation of National Building Code alone is not sufficient, equally important is municipality's structure, procedure, capacity and commitment to safety (ADPC, 2010).

³ According to NBC, buildings types are classified into (A) International states of Art, Engineered (National/international Building Codes), (B) Engineered, (C) Mandatory Rules of Thumb (Pre Engineered), (D) Low strength masonry, Non Engineered

Table 5.1 Collapse of houses in different locations of Kathmandu Metropolitan City

<i>S. No.</i>	<i>House location</i>	<i>No. of storeys</i>	<i>People inside the house</i>	<i>People killed</i>
1	Thado Baryang, KMC 15	4	75	38
2	Sattale, KMC 6	7	90	30
3	Bijeswore, KMC 15	6	80	28
4	Bijeswore, KMC 15	3	350	15
5	Gongabu, Tokha 13	9	200	17
6	Faika, KMC 6	5	100	12
Total				140

(Source: Himal, 2015)

Municipality's system for monitoring building construction is weak and need significant improvement. Analysis of data from survey shows that there is practice of constructing buildings without getting official permission from the corresponding municipality. It also shows that new construction in post-earthquake period are dominated by RCC, but the data show that many buildings do not meet many technical requirements for such structure. Data shows that construction of engineering buildings are very high in peri-urban area, which are highly vulnerable if not constructed properly, therefore, municipality's monitoring and control mechanism should be very high in such areas. It is important note that several RCC frame structure buildings were collapsed in this earthquake causing huge number of death in the area such as Gongabu.

Conversion of Use of Residential Houses

Many residential houses are being converted into schools, colleges, training centers, and even to hospitals and office complexes. Residential buildings are designed for single or multiple families whereas schools, nursing homes, offices and training centers accommodate 'masses' of people. Such changes in use of building irrespective of building structure, type and age is increasing earthquake risk in the valley.

As per prevailing building bylaws, provisions of setback, parking requirements and open spaces including vehicular access are required for the construction of 'mass' gathering facilities such as schools, nursing homes and so on. However, many such mass gathering and critical facilities are being operating in ordinary residential buildings, even without vehicular access. The building bylaws do not have adequate clauses on these matters, which is increasing earthquake vulnerability. In fact, such critical amenities, which should play a crucial role during earthquake emergencies will be the first to collapse in case of a big earthquake.

Chances of converting residential houses into commercial buildings is very in market towns, because only few buildings under construction were found as commercial buildings during survey, majority of them were constructing as residential buildings.

The GoN has attempted to address these issues through recently published directives. Accordingly, anyone wishing to change the use of their building must now obtain permission from their municipality. In order to receive permission, such changes in building use should not be against the building bylaws and national building code. The local municipality can write to the public utility agencies to disconnect them from basic facilities if changes in function are undertaken without permission. Strong monitoring system with efficient enforcement mechanism need to be developed to overcome this issue.

Insufficient Human Resources in Municipality for Reconstruction

The study team visited few municipalities such as Kirtipur, Tokha, Karya Biayak, Bhaktapur, Lalitpur, Madhyapur Thimi to know municipality's human resource capacity. All these municipalities have provision of Architecture (1), Environment Engineer (1), Structure Engineer (1), Civil Engineer (2) in minimum. In addition to these, they also have overseers, sub-overseers, Amin (assistant surveyor) and administrative staffs, to assist the building permit process. These Human Resource Infrastructure is sufficient in normal situation but this may not be sufficient when reconstruction starts. Every municipality need to make Human Resource plan to serve reconstruction phase.

KMC and LSMC have been implementing electronic Building Permit System (eBPS) for faster processing of building permit and maintaining database of the permit issued. Kirtipur Municipality is also in the process of deploying such system very shortly to make building permit system more efficient. This systems in some extent help these municipalities to handle increasing number of building permits for reconstruction. Extension of eBPS in other municipalities can also help them in fast processing with limited number of staff.

Lack of trained masons in building construction

The survey shows there is lack of using trained masons in building construction which is a very critical issues in quality building construction. Therefore, use of trained masons have to reflect in the permit system and government as well as organization working in this sector have to develop strategies so that sufficient number of trained masons can be produced in short time.

Cultural heritage conservation in core settlements and historic towns in fringe

The earthquake destructed areas comprises of various monuments and private houses with heritage values. Community spaces and artefacts such as temples, chityaas, patis (rest place) and dabalīs including procession routes have cultural meanings. Hence, both tangible and intangible cultural heritages must be conserved in reconstruction process. Various proposal of 'house pooling' with widening of road network, changing of community spaces (courtyards, squares, etc.) needs serious discussion among the heritage experts, UNESCO, DOA and local communities including professional bodies. Both the Ancient Monument Act 1976 and Building Bylaws need brick exposed façade, cornices at floor levels, traditional doors and windows and sloped roofs while rebuilding (or renovation) of the houses in the core settlements. The essence of 'newari' architecture and its quantitative aspect has to be maintained, past practices show that such elements are placed on the building facades on ad-hoc basis with compatibility to the surrounding houses.

New construction on small plots – costly, vulnerable and unhygienic

Site visit of the damaged areas, consultation with municipal staffs and KII and FGD in the core settlements and historic towns in fringe areas confirmed that many plots in the core settlement

and historic towns in fringe area are small and elongated due to property division. Layout of rooms and staircases will be extremely difficult in such plots, besides getting adequate natural light and ventilation inside the houses; and foundation footings were not found cantilevered. Reinforce Cement Concrete (RCC) frame structure will also be expensive than normal construction due to needs of raft foundation and minimum column size of 30cm X 30cm (as per National Building Code), irrespective of the span. Moreover, despite engineering construction, such structures will always be vulnerable not only due to the cylindrical shape of the houses but also because of the ‘pounding’ effects against the seismic vibrations. Transportation of building materials and construction management of rebuilding on individual house basis will again be costly and at the same time difficult to regulate the traditional architectural features over building design and façade treatment.

In the core settlement, there has been reduction of habitable spaces due to fragmentation of plots. Even a single house before the earthquake will be rebuilt as many independent houses, depending on the number of property division. Many illegal construction in the form of extra floors and conversion of balcony into room is quite common. However, the post-earthquake reconstruction must be within the height limit (and within FAR) prescribed by the building bylaws.

Land and house ownerships

Land and house ownership issues have also emerged in post-reconstruction and recovery process. Multiple ownership on a single house is a common phenomenon in a ‘Newari’ community due to tradition of living in a joint family and social system of dividing the parental properties equally among brothers. In many instances, family members are having ownership on rooms located at different sides in different floors; others own lobby and staircase areas. Such ownership is often fixed on mutual understanding among family members rather than through legal documents. The worse situation is that some owners have sold their rooms (and other spaces) to the third party, other than their family members. Consensus among them in sharing spaces (rooms), style of new building and timing of construction and investment to be done in post-earthquake construction has become a daunting job. Those victims living in temporary shelters prefer to rebuild the house at the earliest possible whereas other owners having houses in other places are less interested in investing in the collapsed houses. There is a conflict and debate regarding ownership over the ‘common walls’ between two houses, as ‘Newari’ houses are often seen in compact form with common walls at least in one side. In such a jointly built houses, demolition of the damaged part can also lead to destruction of the surrounding structure.

Recovery of poor renters and informal settlement

Among various communities, the poor renters are the most affected, and their condition is measurable. Even before the earthquake, they had been renting unhygienic spaces in inadequate light and ventilation old houses of the core settlements, as they used to get job in nearby neighbourhoods. They have become the victims of the earthquake from different fronts. They not only lost their jobs but their renting houses also damaged. They need to pay high rent in other location, where they also need to find new jobs. This group of people are still living in damaged and vulnerable houses. Reconstruction of those houses will definitely displace them again, as they will not be able to pay the high rent. Only house owners were eligible to get the government’s immediate financial support of NRs, 15,000 and another NRs. 10,000. Those living in the temporary shelters also have similar problems.

Identifying the poor renters is very difficult task in urban areas due to unavailability of proper data and information. Municipalities either do not have a good system to record the renters in the city. Municipalities must have to develop a mechanism to maintain database of renters in the city, so that who and where about of the renters can be known to for their livelihood.

Vulnerable dense urban fringe

Dense settlement like 'Ghatekulo' was little damaged by the Gorkha earthquake. Illegal extra floors, irregular and haphazard pedestrian network and above all lack of usable public open spaces have caused panic to households in these areas. They spent many days at Tundikhel or some other locations for safety for many days after the earthquake, as there was hardly any safer open spaces within their neighbourhood.

Need to revisit high rise apartment in the Kathmandu valley

Though no major structural damage did not occur on high rise apartments by the earthquake, there are losses of individual unit owner due to damages on interior, which is not covered by the insurance. These people managed alternative accommodation either renting spaces in other places or adjusting at relative's homes for about six months before returning to the apartments. Due to high rise structure, most of the open spaces of the apartment premises could not be used for temporary shelters after the earthquake. According to some unit owners during FGD, they could not have returned to the apartment if they could have other options and would like to dispose if buyers available. They are ready to sell their units upon getting buyers. Market value of the units have not improved in the real estate market. In the period of energy crisis, it was inconvenient to walk all the way to higher floors without lift service. After the earthquake, many people have not developed positive connotation towards high rise apartment. In nutshell, it has created a debate high rise apartment versus low rise high density development.

Focus on individual house rather than settlement scale

The National Reconstruction Authority (NRA) has been assigned huge responsibility ranging from preparation of plans (integrated housing, house pooling and relocation of vulnerable settlements) to formulation of necessary bylaws and regulations to implement them including monitoring and supervision of sites as per Earthquake Affected Infrastructure Reconstruction Act 2015 (2072 BS). However, various governmental departments under different ministries and the concerned municipalities (as per Local Self-Governance Act 1999) are also responsible for urban (re)development activities. On the top of that, numerous international nongovernment organisation, donor agencies and local community organisations have also been involved in reconstruction works. Hence, coordination and cooperation of NRA with these agencies to avoid duplication of works and their jurisdiction areas is the pre-requisite condition for the success. Need of sustained redevelopment on inclusive basis have been cited in the directives by the Ministry of Federal Affairs and Local Development (MoFALD) and Ministry of Urban Development (MoUD) including in the Earthquake Affected Infrastructure Reconstruction Act 2015. However, various activities carried out on the ground have basically focused on physical aspect with little linkages to improvements of livelihood and business activities. Also, the boosting of construction industries and building material supply is also yet to get priority.

Inadequate regulations for implementation of reconstruction plan

The two ministries' namely MoFALD and MoUD's directive clearly spelt out the concept of 'house pooling' and 'urban regeneration' for reconstruction of the core settlements and historic towns in fringe areas. They have assigned the concerned municipalities for preparation of necessary regulations and guidelines before executing such projects. As per Infrastructure Reconstruction Act 2015, NRA can also do this job. So far, municipalities in these study areas are busy in debris management, revising building bylaws and issuing building permits including damage assessments. Almost all of them are yet to come up with redevelopment and reconstruction plan and program along with regulations and guidelines towards house pooling and urban regeneration. Moreover, there are confusion regarding reconstruction plan and recovery on the earthquake damaged areas in two ways. First, it is not clear which agency is responsible for giving planning permit in the newly planned earthquake damaged areas. There are many agencies working for reconstruction and recovery: NRA, MoUD, KVDA and MoFALD. Second, due to lack of clear cut guidelines for reconstruction in the urban damaged areas, no one is sure if the proposed plans comprises of changes in the form and size of open spaces, street width and even building footprints. If the small individual houses are replaced by a single unit with mixed uses, what are the provisions for distributing (and giving compensation) among the participating households are not clear. Also, does the concerned municipalities issue building permit for proposal that combines three to four plots into one single unit with common staircases? If the existing 'Ownership of Joint Apartment Act, 2056 (1997)' is applied, it requires not only a separate procedure for getting planning and building permit but also needs different provisions on building density and setback requirements. As a result, some of the community initiation carried out in different parts of the valley has been limited to the paper only.

Financial aspect

The GoN's has announced either giving a grant of NRs. 200,000.00 (two lakh) with concessional loan up to NRs. 300,000 or providing a soft loan up to NRs. 1500, 000 for outside Kathmandu valley and 2500,000.00 (twenty five lakh) for the valley. Banks and Financial institutions (BFIs) will get refinance (5-10 years) facilities at 0% interest for a year and renewed thereafter. BFIs cannot charge more than 2% interest except insurance, collateral valuation, credit information and guarantee fee. However, majority of victims are not aware on complete procedure and policy. Loans disbursement against income of client and collateral will be difficult for victims, as they lost their income and assets in the earthquake. The refinancing facilities are limited for only 10 year period whereas other housing loans are available for 15 year as medium term and 25 years as long term. As only those who do not have other house than destroyed one is eligible, many households of the destroyed houses will be out of this scheme. The total loan amount under the scheme will be a maximum of 80 percent of the core capital of a bank or financial institution. The loan will be treated as a home loan and carries a risk weightage of 60 percent as in other home loan for BFIs. Loan loss provision for this insured home loan will be 25 percent of standard norms prescribed by NRB. BFIs have to submit monthly report of this refinance loan scheme to NRB.

Resilient society

Despite loss of relatives and properties and need to stay in temporary shelters, people in the study areas have been found resilient. During the survey period, they are not only cooperative but also offer tea and also invited to visit their places. The family structure, social bonding among relatives and neighbourhoods including cultural framework through performance of various rites and rituals have made this possible. Moreover, while staying victims in one place,

they were able to share their grievances and difficulties, which also helped them to be calm even in such natural disaster.

Community mobilization

Many communities in the core settlement and historic towns in fringe areas have come up with their own redevelopment plan. About thirty five households at ward no. 16, Kaminani tole, Tekhachen in Bhaktapur municipality has formed a community group since the gorkha earthquake. This group has been found very active in rescue and relief operation. The community has also worked with a professional team (from Japan and Nepal) under J-Rapid project for the reconstruction and recovery of their community at the earliest possible. After series of consultation, discussion and community workshop, they have selected one proposal (out of three proposed), which suggests to combine small plots (3-4 plots into one site) for the construction of buildings due to its multiple advantages. Similarly, 82 households of 'Newar community' at Pilachhen in Lalitpur sub-metropolitan city has come up with a proposal of reconstruction with provision of guest house and galleries on the lower floor with residential uses (for house owners) on the upper floors in the newly built houses. The community with support from private sectors also agreed on the financial mode of reconstruction: 25% cash payment by individual house owner, 25% cash in kind support, 25% volunteer support and 25% from bank financing. In this proposal, each individual will build a separate building in its own plots, without combining the plots and without changing the open space and street width. Another 80 household of Kilagal in Kathmandu metropolitan city also agreed to participate in the house pooling scheme with support from NSET. Accordingly, new plan has been prepared by replacing the small houses by bigger units with provision of commercial activities on the lower two floors with provision of flat system housing units on the upper ones. The earlier form and shape of the open spaces (courtyards) and pedestrian lanes have been changed and replaced by new form in different locations. In Khokana too, the local community has prepared the reconstruction and recovery plan with detailing of the streetscape. Many confusion exist for implementation of these proposals: what and how to conserve cultural heritage and building typology in these old settlements; who are the right agency to get the permission; does planning permit is necessary for such proposal and so on are yet to be answered. Nonetheless, the motivation of local communities for building up their neighbourhoods and towns is a very positive initiation, which needs to be realised with support from public and financing agencies.

5.2 Long Term Impacts

Unlike in other countries, where earthquake victims are kept in temporary housing, the earthquake affected households in the core area are living in temporary shelter. Hence, building their housing units is urgent and must get the top priority in national reconstruction process. Delay in reconstruction and recovery process as well as present trend of response may lead numerous long term implications.

Opportunity loss of integrated planned development

Redevelopment strategy focusing on individual house basis means it misses many opportunity of rebuilding safer neighbourhoods and earthquake resilient community. Not only individual houses but equally important is the infrastructure (physical, social and emergency facilities) including transportation (pedestrian and vehicular) network and open spaces must be planned in an integrated way not only for disaster preparedness but also for rescue and relief operation.

Difficult for achieving safer and cost effective as well as conservation of cultural heritage

The reconstruction of individual houses in the core settlements and historic towns in fringe will not be safer, cost effectiveness and above all it will be difficult to conserve the liveliness and architectural heritage. Majority of historical and heritage private houses may vanished because of unaffordable maintenance and reconstruction cost affecting potential tourism industry. It takes a longer time for rebuilding. It will be significant impacts on longer terms. It will dilute the earlier status of World Heritage List thereby decreasing the tourists' flow, which will have negative impacts on local handicrafts industry and income of the municipalities. As a result, it will reduce the income and affordability of local communities relying to tourism related businesses. In the long run, they need to move to other places for business and income generation thereby impacting the intangible heritage – celebration of festivals, rituals and so on. In nutshell, the present built environment will be slowly converted into concrete junction. Illegal construction even in post-earthquake period particularly in the core settlements and historic towns in fringe areas will likely take place to give habitable spaces for the increasing family members. As significant percentage of spaces will be consumed for circulation and staircases while designing houses in small plots in the core areas, there is no option for the household in present situation (as per building bylaws and lack of integrated housing schemes) other than going for illegal construction. Such structures possess risk not only to occupants but also neighbourhoods and pedestrian around the houses. Monitoring of such structure on individual basis will be difficult for the concerned municipalities, as demonstrated well in the pre-earthquake period.

Conversion of temporary shelters into slum areas and formation of new squatter settlements

Chances of converting temporary shelters into slums is likely to take place. This is possible due to delay in reconstruction and high cost of new construction, which is beyond the affordability level of many victims. Poor renters currently residing in vulnerable structures may not be able to afford the high rent in newly built houses. Staying in temporary shelter for many years means those occupants must have arranged their business and income generating activities in the nearby vicinity. Removing such shelter in future will be complicated and it will also hamper their job and income. Forceful removal of them may create new squatter settlements along the riverfronts and other vulnerable sites.

6 The Way Forward

Numerous organisations including Government of Nepal are working hard for recovery and reconstruction of earthquake damaged areas. Local communities are very much committed on it. Some tangible progress are visible particularly the policy towards building safer neighbourhoods and earthquake resilient communities in the reconstruction process. However, to realise the broad level government's policy into local level, there still some gaps. The present approach of reconstruction at individual house scale shall be replaced by settlement level integrating infrastructure provision and housing construction on a holistic way. Physical redevelopment shall be combined with livelihood upgrading for the sustained long term recovery. It will boost up their moral confidence, improve their quality of life and help to pay back loan in time. Safer and cost effective redevelopment together with conservation of cultural heritage in the core settlements and in historic towns in the fringe areas will be possible on in integrated approach. Moreover, the opportunity to incorporate the present emerging issues such as energy efficient components (rain water harvesting, solar panels, etc.) is possible through integrated infrastructure provisions linking with construction of housing units. Various components of disaster preparedness as well as planning for emergency situation in rescue and relief operation shall be planned in such integrated approach. The construction cost will be down and quality control during construction period will be effective.

In such approach, though there are individual private houses, inclusion of integrated infrastructure plus planning of total built environment thereby conserving cultural heritages (neighbourhood environment, social and cultural spaces, etc.) as well as promoting tourism development. Hence, many donor agencies, (I)NGOs and public utility providing agencies can also easily participate by providing financial and technical support for the planned integrated development. The local community feel indirect support in funding (and financing) infrastructure by these agencies. Moreover, early involvement of business entrepreneurs, real estate companies, building material suppliers will not only further ensure the use of newly developed spaces but it will also confirm continuous supply of materials and quality control. In such situation, local communities will be further motivated to involve actively in the reconstruction process thereby taking its ownership. The existing regulatory mechanism must be simplified focusing municipality as one single window for getting planning and building permit, loan (or grant) disbursement and construction supervision and monitoring. Such collaborative approach will ensure fast reconstruction. As a result, significant number of households staying in vulnerable structures can be easily housed in the safer neighbourhoods. It will not only fulfil the basic goal of active community participation in rebuilding in post-earthquake period but it will also strengthen community bond and self-help practice, which ultimately make the community more disaster resilient in future.

Based on above mentioned strategy and approach, a theoretical recovery and reconstruction model has been developed. It is a matrix (3 X 5) comprising of three columns – major activities at the centre, supported by actors on the left and mild stone of the reconstruction phase on the right side (Fig. 6.1). Similarly, there are five distinct redevelopment phases from concept to completion and post-disaster management. This model covers not only the collapsed and damaged areas but it also deal with other vulnerable structures, which need retrofitting and structural treatment.

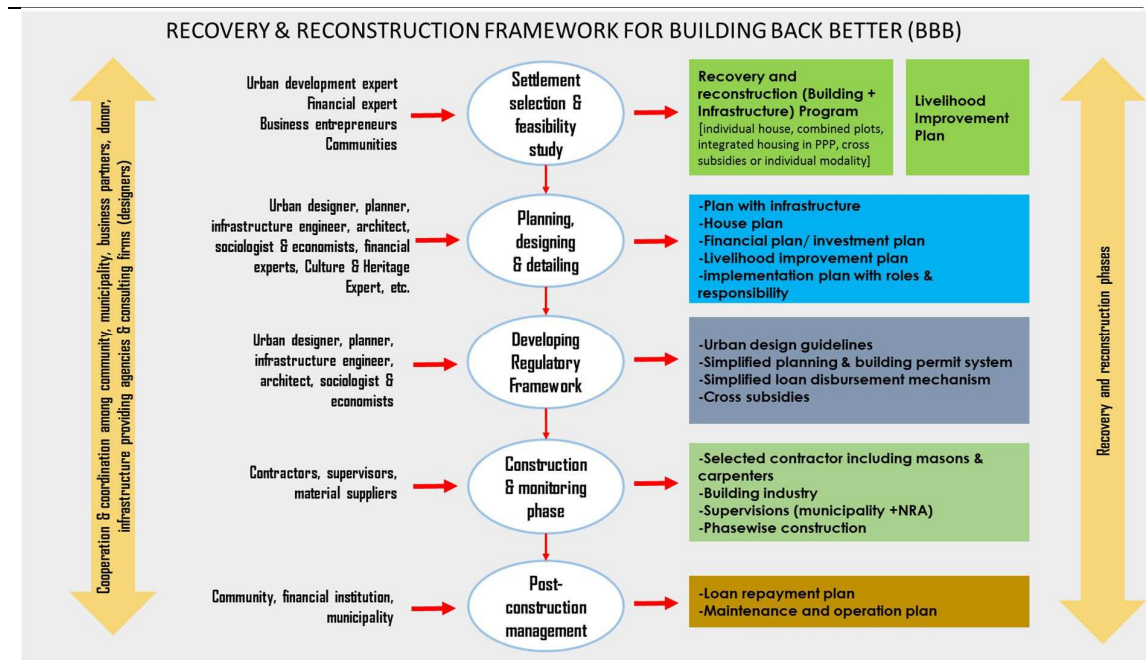


Fig. 6.1 Recovery and reconstruction model based on integrated development planning

Site selection and feasibility study of mode of reconstruction (PPP, government subsidies, individual contribution, etc.) shall be carried out in the initial stage with teams of experts in the field of urban development, finance, business entrepreneurs and the community itself under the guidance of NRA and the concerned municipality.

Once the mode of redevelopment and the financial modality is agreed in principle, the detailed planning of the damaged area together with layout of housing units with construction detailing shall be prepared with team comprising of urban designers, infrastructure engineers, architects, sociologists, financial expert, Cultural & Heritage Expert, and so on. Consortium of such experts shall critically reviews various causes of failures of both unreinforced masonry structures (in the core settlement and historic towns in the fringe) as well as RCC structures (in peri-urban and market towns) and come up with earthquake resilient space planning, construction detailing and selection of building materials. Not only for reconstruction of new structures, but retrofitting of the existing vulnerable structures is also necessary. The expert team should also come up with effective solution for this problem. This stage is most important part of the model. First, various plans such as physical infrastructure development, layout of rooms in each unit in detail (architectural drawing and working detailing), investment plan, flow of funds and finance, roles of various stakeholders, and so on must be prepared and finalised with intensive discussion with those concerned agencies. Financial intermediary such as Town Development Fund (TDF) can provide soft loan for infrastructure development. In addition to these, financial contribution and flow of the fund shall be finalised. Small plots can be combined with common staircases, which will have multiple advantages: increase in habitable spaces, stable building form, cost effectiveness. Communities need to be awarded in terms of cost and space effectiveness on differences between building individual houses on small plots and combined plots. The role of local communities and legal expert of the concerned municipality will be crucial in this stage to tackle down the issues associated with multiple ownerships on house and lands. Local influential leaders and social workers can play a catalytic role in managing issues among families and neighbours on land and house ownership. Second, various design guidelines shall be developed not only to ensure

conservation of cultural heritages but also to incorporate present day needs (energy efficient components) so that not only those participated in the project but also those who individual who do not participate in the damaged areas can be benefitted. It will also help to simplify the building permit regulatory system. Third, various committees with active involvement of the earthquake victims should be formed for transparency of the budget and smooth monitoring of the construction work.

The third stage, getting planning and building permit from the municipality is another mildstone of the model. Besides municipality, NRA and DOA (in the core settlements and historic towns in the fringe areas) will also be involved in loan disbursement as well as ensuring the provision of Ancient Monument Act 1956. As getting building permit on individual basis dealing with various government agencies might be time consuming, it is suggested to make this process simplified and shall be dealt with one door policy, i.e., from the concerned municipality only. Other government agencies can coordinate and collaborate with municipality, whose main function is to issue building permit, monitor construction supervision including fee collection. In order to facilitate reconstruction work fast and effective, planning and building permit system shall be carried out on project basis, instead of individual house cases.

Private sector (real estate companies and building material suppliers) collaborate is suggested for the quality and timely completion of the reconstruction work. However, all the houses in the damaged areas need not to be reconstructed by hiring such companies. Individual household having bigger plot area, where single independent house construction is possible can manage the construction work by themselves. The local masons and carpenters shall be involved wherever possible. In this stage, on-site training for masons and carpenters can be combined for human resource development.

The final stage, i.e.' post-earthquake management is crucial for operation and maintenance of the structure as well as for loan repayment on time.

In order to support the ongoing initiation of recovery and reconstruction process, the following activities are suggested for immediate actions.

- (a) Feasibility study regarding mode of development shall be carried out at different locations – core settlement, historic towns in fringe, peri-urban areas and market towns based on settlement pattern, affordability of the occupants, business opportunity and so on through consultation with earthquake victims, local business entrepreneurs and development experts;
- (b) Identification of historical planning and design principles, architectural typology and craftsmanship including cultural practices (UNESCO's criteria for declaring WHS) that need to be conserved in the core settlements and historic towns in the fringe. Formulation of urban design guidelines to protect the shared community values as well as inclusion of energy efficient components is essential to guide architects, municipal engineer and other design professions. It should also include guiding principles for construction on plot and house combination.

- (c) Development of a mechanism to simplify planning and building design permit for house owners. Such mechanism should also ensure effective roles of different government agencies – NRA, DOA, Municipality and infrastructure providing agencies – in plan preparation, issuing permit, loan disbursement including monitoring and supervision of construction works;
- (d) A financial model that also include household awareness should be developed so that each household will feel comfortable in contribution towards joint construction, loan repayment and so on.
- (e) A comprehensive study need to carry out to know the where about of the poor renters and their living status after earthquake and develop a plan for their shelter and livelihood.
- (f) Incorporate informal settlement management plan into government’s normal planning process to improve the livelihoods of people living in the informal settlements.

The concerned municipality being an implementing agency should play a central role. As the municipalities lack expert human resources on urban design, conservation, infrastructure planning, finance and legal, MoUD and DUDBC can provide technical support, if necessary shall be hired. NRA can also play a constructive role not only by brining all the involved actors (public, private, NGOs as well as designers and local communities) under one platform but also simplifying the existing rules and regulations.

Bibliography

- ADPC. (2010). *Urban Governance and Community Resilience Guides. Mainstreaming Disaster Risk Reduction*. Asian Disaster Preparedness Centre, 2010 A.D.
- Awale, S. (2014). *The Himalayan Time*. Local daily, 15 August 2014, A.D
- CBS. (2012). *National Population Census 2011, Household and Population by Sex Ward Level*. Central Bureau of Statistics, November 2012 A.D.
- CBS. (2014). *National Population and Housing Census 2011, Village Development Committee/Municipality, Volume 06, NPHC2011*. Central Bureau of Statistics, National Planning Commission, Government of Nepal, March 2014 A.D.
- CBS. (2016). *Four monthly Statistical Bulletin (Year 37; Vol. 112/113, No. 2/3)*. Central Bureau of Statistics, Kahtmandu, 2016 A.D.
- Census. (2011). *Population Census 2011, Central Bureau of Statistics, National Planning Commission*. Government of Nepal, 2011, A.D, Website: cbs.gov.org.np.
- DOA. (2015). *Preliminary Report of Monuments affected by Earthquake April 25*. Department of Archaeology, 2015 A.D.
- Gutschow, N. (1993). *Sacred patterns of a living urban tradition*. In: Spodek, H. and Srinivasan, D. M. ed., *Urban form and Meaning in South Asia: The shaping of cities from prehistoric to precolonial times*. Hannover and London: University Press of New England, 1993 A.D.
- Himal. (2015). *Great Earthquake 2072 BS. Year 24. Vol.6. 24-30 May*. Himal Weekly Magazine (in Nepali language), 2015 A.D.
- HMG. (1976b). *Kathmandu Valley – the Preservation of Physical Environment and Cultural Heritage Protective Inventory, Vol 2*. Department of Housing, Building and Physical Planning, His Majesty's Government of Nepal, 1976 A.D.
- HMG. (1996a). *Kathmandu Valley – the Preservation of Physical Environment and Cultural Heritage Protective Inventory, Vol 1*. Department of Housing, Building and Physical Planning, His Majesty's Government of Nepal, 1976 A.D.
- Hosken, F. (1974). *The Kathmandu Valley Towns*. New York: Weatherhill, 1974 A.D.
- HRRP. (2016). *HRRP News Letter, Vol. 1, Dec 2015 – Mar 2016*. Nepal Housing Recovery and Reconstruction Platform, 2016 A.D., Website: www.hrrpnepal.org.
- Joshi, H. (2060). *Festivals of Nepal*. Joshi Research Institute, Patan, 2060 B.S.
- KMC/ICIMOD. (2015). *Inventory and documentation of private heritage homes and public monument of Kathmandu*. Kathmandu Metropolitan City, 2015 A.D.
- Malla, M. (2016). *Festivals celebrated by Newars of Kathmandu Valley*. Nepal Heritage Society, Bhadrakali, Kathmandu, 2016 A.D.
- Malla, U. M. (1978). *Settlement Geography of Kathmandu Valley*. Geographical Journal of Nepal. (1): 28-36.

- Muzzini, E. and Aparicio, G. (2013). *Urban Growth and Spatial Transition in Nepal*. An Initial Assessment. Washington, The World Bank, 2013 A.D.
- NRA. (2015a). *Procedures for the Flow of Grant/Assistance for Reconstruction of Houses Completely Destroyed by the Earthquake*. National Reconstruction Authority, 2015 A.D.
- NRA. (2015b). *Procedures for Mobilizing the Non-Government Organizations in Reconstruction and Rehabilitation work*. Kathmandu Nepal: National Reconstruction Authority.
- NRB. (2011). *A Report on Real Estate Financing in Nepal: A Case Study of Kathmandu Valley*. Nepal Rastra Bank 2011 A.D.
- NRB. (2015). *Procedures for Refinancing to be provided for Reconstruction of Residential Houses by affected by Earthquake*. Nepal Rastra Bank, 2015 A.D.
- NRB. (2016). *Nepal Rastra Bank Samacharpatra April-May 2016*. Nepal Rastra Bank, 2016 A.D.
- NTB. (2011). *Cultural treasures of Nepal with special focus on Kathmandu Valley, Nepal Tourism Year*. Kathmandu, Nepal, 2011 A.D.
- Oldfield, H. A. (1974). *Sketches from Nepal*. New Delhi: Cosmo Publication, 1974 A.D.
- PDNA. (2015a). *Post Disaster Needs Assessment, Vol. A, Key Findings*. National Planning Commission, Government of Nepal, Kathmandu 2015 A.D.
- PDNA. (2015b). *Post Disaster Needs Assessment, Vol. B, Sector Report*. National Planning Commission, Government of Nepal, Kathmandu 2015 A.D.
- Pilachhen. (2015). *The Pilachhen Reconstruction and Tourism Promotion Project*. Promoted by Maya Foundation, Pilachhen, Lalitpur, 2015 A.D., Website: www.mayafoundationnepal.com.
- Regmi, D. (1965). *Medieval Nepal. Part I*. Calcutta: P.L. Muckhopadhyaya, 1965 A.D.
- Shrestha, B. K. (2011). *Street typology in Kathmandu and street transformation, Urbani izziv*, Vo. 22, No. 2, pp. 107-121.
- Shrestha, C. B. and Vaidya, K. L. (n.d.). *Settlement Patterns in the Kathmandu valley. The Himalayan Review*. Nepal Geographical Society, Vol. 10.
- Slusser, D. R. (1982). *Nepal Mandala (1)*. New Jersey: Princeton University Press, 1982 A.D.
- Wolfgang, K. (1976). *The Traditional Architecture of the Kathmandu Valley*. Kathmandu: Ratna Pustak Bhandar, 1976 A.D.